

Index Option Trading Activity and Market Returns

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Abstract

Do order flows in index derivatives play an informational role? Weekly index put order flow on the International Securities Exchange positively and robustly predicts weekly S&P 500 index returns. This result obtains mainly for net put buying and is stronger in high VIX periods and in periods following macroeconomic announcements. We explore rationales for our findings, which include investor sentiment, the notion that market makers trade on information in options markets, and option-based risk protection strategies used by retail investors. The last explanation accords best with our analysis.

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1. Introduction

Options markets have become increasingly important in the last several decades. There are at least two reasons for their success. First, options enhance welfare by completing markets (covering more contingencies); viz. Ross (1976). Second, as Black (1975) points out, options provide leverage and thus could allow agents to trade more effectively on their information. Easley, O'Hara, and Srinivas (1998) find that options order flows contain information about the future direction of the underlying stock prices. Cao, Chen, and Griffin (2005) find that options volume predicts equity returns around takeover announcements, which accords with informed trading in the options market prior to corporate events. These studies suggest that informed traders prefer options to individual stocks when option implicit leverage is high and when options are relatively liquid. Pan and Poteshman (2006) find that put-call ratios predict future individual stock returns. Other studies also show that option prices can predict future stock returns and contribute to price discovery.¹

The preceding studies principally focus on individual stocks. In contrast, this paper focuses on *index* options. While private information is, *prima facie*, less relevant for the overall market relative to individual companies (Gorton and Pennacchi, 1993), Bernile, Hu, and Tang (2016) and Kurov, Sancetta, Strasser, and Wolfe (2019) do find evidence of informed trading in stock index futures markets prior to macroeconomic announcements. However, evidence of informed trading in index options remains scant. For example, Pan and Poteshman (2006) do not find evidence that net buying in puts and calls predicts returns in S&P 100 (OEX), S&P 500 (SPX), and NASDAQ

¹ Chakravarty, Gulen, and Mayhew (2004) show that options account for a significant share of price discovery in the underlying stocks. Cremers and Weinbaum (2010) show that the difference between implied volatilities of call and put options on the same stock with the same strike and time to expiration (the “volatility spread”) contains information about future returns. An, Ang, Bali, and Cakici (2014) also show that option implied volatilities forecast the cross-section of stock returns.

100 (NDX) index options. In contrast to Pan and Poteshman (2006), we use a unique data set from the International Securities Exchange (ISE), which contains the complete daily record of buy and sell activity in index options over a twelve-year period, together with details on whether a transaction is involved in opening or closing an options position. These options are actively traded; indeed, on the ISE, the notional volume in index options is about one-fifth of the total notional volume in all individual stock options during our sample period.

We compute the order imbalance (OIB) of stock index options as the difference between weekly position-opening buy and sell trading volumes (in number of option contracts), divided by the total weekly volume of position-opening option trades. We examine whether index call and put option OIB (or signed order flows) convey information about stock market returns in subsequent weeks. Order flows are widely used to study informed trading. Evans and Lyons (2002), Beber, Brandt, and Kavajecz (2011), and Menkveld, Sarkar, and van der Wel (2012)² argue that order flows can affect price formation beyond past returns whenever the transfer of informational signals from order flows to current prices is imperfect, possibly owing to the fact that order flow is not perfectly observable in real time. Consistent with their view, we find a strong and statistically reliable predictive ability of signed ISE index options order flow for weekly index returns.

The predictability we document survives controls for macroeconomic variables, lagged index returns, and lagged order flows from other index derivatives, within a vector autoregression (VAR) framework. The results are economically significant; impulse responses from the VAR indicate that a one-standard-deviation shock to index put option order imbalance forecasts a

² These papers examine the information role of order flows in the context of currencies, industry sectors, and Treasury markets, respectively.

cumulative shift in the S&P 500 return of about 0.55% over the next ten weeks. This amounts to an annualized return of about 2.9% per year.³

We find that net buying of ISE put options by customers positively predicts index returns. The predictability is stronger during high VIX periods. The predictive power of put options survives in the post-crisis period, indicating that our result is not an artifact of the financial crisis. To isolate the role of options order flow relative to that in other derivatives and the index itself, we include order flow from alternative index instruments in the VAR. Specifically, we include the order flow in the underlying constituent stocks, an index futures market, and the ETF on the S&P 500 index. The return predictability of index put options survives all these inclusions.

We consider possible rationales for our results. First, return predictability may reflect “negative” sentiment of investors. For example, option customers may wrongly bet that market returns will be high when investor sentiment is high and expected returns are low.⁴ The results do not accord with a sentiment-based explanation because the inclusion of sentiment proxies in our vector autoregressions does not alter the predictive ability of put option order flow for index returns.

Second, since net buying of puts by customers is tantamount to net put selling by market makers, and such order flow positively predicts market returns, option market makers may have information relevant for predicting market returns. Indeed, market making firms in options markets are large institutions such as Citigroup, Morgan Stanley, Goldman Sachs, Citadel, etc.,⁵ while the customers are mainly retail traders.⁶ We do find that the predictive power of put options

³ This is comparable to the annual magnitude of 7.7% of the value effect in Fama and French (1998). Their finding applies to individual stocks, whereas in our case the predictability is for the S&P 500, a highly liquid and visible index.

⁴ Barber and Odean (2000) find retail investor trades negatively predict individual stock returns.

⁵ See <https://web.archive.org/web/20160613064847/http://www.ise.com/options/membership/exchange-members/>

⁶ For example, Kurov and Lasser (2004) and Osler, Mende, and Menkhoff (2011) provide evidence that markets makers in index futures and currency markets contribute to price discovery. Similarly, Anand and Subrahmanyam (2008) show that equity market intermediaries contribute more to price discovery than other traders.

order flow for index returns is stronger during weeks with major (pre-scheduled) macroeconomic announcements. But the differential predictability obtains in the week of and the week following the announcements, not prior to the announcements. Further, an upward movement in bid and ask quotes, which implies that informed market makers encourage sells because they have positive information, is not accompanied by higher put returns. These findings do not support the information hypothesis.

Another interpretation of our results is that agents buy put options as insurance during periods of increased uncertainty, so that put buying signals higher market premia, i.e., higher required returns.⁷ Supporting this explanation, the predictability from put order flow obtains principally from net put buying and is higher during high VIX periods. Further, innovations to put order flow are positively correlated to VIX innovations, suggesting higher demand for puts when uncertainty (as measured by VIX) is high. Our analysis thus indicates that market makers provide insurance to outside investors when uncertainty is high.

On balance, the evidence supports the insurance explanation, as opposed to the explanation relying on informed market makers. It is intriguing, however, that while predictability of index returns from ISE index options order flow is strong we find that order flow from SPX options on CBOE does not yield predictability. The ISE index options that we study are particularly popular among retail investors, while most of order flow in SPX options is institutional. Indeed, we find that the average trade size is six times smaller for ISE index options than for SPX options. Overall, the popularity of ISE options among (likely uninformed) retail investors potentially reflects the notion that more risk averse retail investors have a greater demand for protection via puts.⁸

⁷ This hypothesis is consistent with Grossman and Zhou (1996) and Bates (2008). In their models, less crash-averse agents insure the more crash-averse agents through options.

⁸ Haddad and Muir (2018) provide a setting where intermediaries are less risk averse than individual investors.

Consistent with this, we find that most of the return predictability is driven by small customer orders. We also confirm that the ISE order flow predicts returns for the corresponding underlying indices, which is not unexpected, as these indices are highly correlated with the S&P 500 index.

In a closely related and important paper, Chen, Joslin and Ni (2018) show that high buying activity in deep out-of-the-money S&P 500 index put options predicts low monthly market excess returns. Their horizon and direction of prediction is different from ours, as we find that net put buying positively predicts market returns at the weekly horizon. The idea in Chen, Joslin, and Ni (2018) is that low put buying implies that intermediaries are not willing to readily supply liquidity to facilitate trading. This is indicative of intermediary constraints, which, in turn, implies higher required returns. Our finding that SPX options order flow is not able to predict index returns at the weekly horizon, and the fact that our direction of prediction is opposite to theirs, suggests that our result captures a different phenomenon, so that the two papers are complementary.

2. Data

The sample period is from January 2, 2006 through December 29, 2017. In the analysis that follows, we use a weekly measurement frequency. The sample period compares to the three-year sample period from 2008 to 2010 used by Hu (2014), and the eleven-year period of Pan and Poteshman (2006), for examining the relation between stock option order flow and individual stock returns. Our choice of the sampling interval is inspired, in part, by Pan and Poteshman (2006), who show that it takes about a month for individual stocks to adjust to news contained in index option trades. This suggests that daily or intraday intervals would be too short for our study. On the other hand, we would expect the overall index (which is weighted towards the liquid, large caps) to adjust to information in order flows more quickly than individual stocks, so considering

intervals of a month or longer might obscure the informational role of *index* options. Hence, we use weekly observation intervals as an intermediate choice.

The structure of the U.S. options market is similar to that of the equity market but has some distinct features. Options are typically cross-listed across multiple fully electronic exchanges, and the NBBO rule is enforced. Investors can post limit or market orders, and market-makers are obliged to provide continuous two-sided quotes. All major brokers provide real-time option prices to their (retail) clients similarly to the way they provide stock information. Muravyev (2016) provides further details on the options market structure.

While we use options order flow in the initial part of the paper, later we also use order flow from other contingent claims as controls. Here are the variables used in the paper.

1) *Order imbalances for index call and put options.* The order imbalance (OIB) of stock index options traded on the ISE is computed as the difference between weekly position-opening buy and sell trading volumes (in number of option contracts), divided by the total weekly volume of position-opening option trades.⁹ The OIB is computed separately for call and put option contracts by combining data for the Russell 2000 Index (RUT) and the Nasdaq-100 Index (NDX). The data are obtained from the ISE Open/Close Trade Profile.¹⁰ Over our sample period, volume in these two index options amounts to about 94% of the total volume of all index options traded on the ISE.¹¹ Notably, our ISE data clearly identify which side of a trade

⁹ We find that position-closing trades do not predict index returns. This is consistent with Pan and Poteshman (2006), who show that option trades initiated to open new positions contain more information than position-closing trades.

¹⁰ A description of the Open/Close Trade Profile data is available at <https://business.nasdaq.com/intel/GIS/ISE-Open-Close-Trade-Profile.html>.

¹¹ The RUT and NDX options were by far the two most actively index options on the ISE during our sample period. The RUT index option was delisted on April 22, 2013, therefore, for the remaining part of the sample we work with the NDX index options. We have verified that including trades from other index options (for example, those on the Russell 1000 and S&P Mid Cap 600 indices) makes virtually no difference to our results. Further, we discuss in Section 4 that including order flow in the S&P 500 (SPX) index options traded on CBOE as a control also leaves our results substantially unaltered.

is taken by option market makers. Thus, we directly observe OIB instead of estimating it from public transaction data by assuming that market-makers never cross the spread and applying the Lee and Ready (1991) algorithm, which may lead to estimation errors. The ISE data is missing for periods from February 8 through February 26, 2016 and from June 5 through July 28, 2017. We winsorize the weekly option OIBs at the first and 99th percentiles.

If moneyness is defined based on the ratio between the underlying and strike prices as $0.9 < S/K < 1.1$ for at-the-money (ATM) options and otherwise for out-of-the-money (OTM) or in-the-money (ITM), then about 2/3 of the volume for these index options is in ATM options, 1/3 in OTM and only about 1% in ITM options. The volume-weighted time to maturity is about 30 days for calls and about 32 days for puts. As mentioned above, our option order imbalance variables are based on position-opening buy and sell volumes. In contrast, Pan and Poteshman (2006) construct their main information variable (the put-call ratio) using only position-opening buy trades. This difference in computation results in the correlation between the weekly put order imbalance and the put-call ratio constructed using the ISE index option data of only about 0.11.

- 2) *Index futures order imbalance.* The weekly index futures order imbalance is computed as the difference between weekly buyer- and seller-initiated volumes (in number of contracts) of the E-mini S&P 500 futures, divided by the total weekly trading volume. Using the trade-by-trade data obtained from Tick Data, Inc., we classify the trading volume as buyer- or seller-initiated using the tick rule. Specifically, the trade is classified as buyer-initiated (seller-initiated) if the trade price is higher (lower) than the last different price. We then aggregate the buyer- and seller-initiated volumes within each week.

- 3) *Stock order imbalance*. The weekly order imbalance for S&P 500 stocks traded on the NYSE is computed as the difference between weekly dollar buyer- and seller-initiated volumes, divided by the total weekly dollar volume. We sign trades as buyer- or seller-initiated using the Lee and Ready (1991) algorithm. The transactions level data is obtained from the NYSE Trade and Quote (TAQ) database.
- 4) *ETF order imbalance*. Using TAQ data for the SPDR S&P 500 ETF (SPY), we compute the order imbalance as the difference between weekly dollar buyer- and seller-initiated volumes, divided by the total weekly dollar volume.
- 5) *Aggregate stock market return*. We use the weekly continuously compounded (log) return on the S&P 500 index, computed using the opening value of the index on the first trading day of the week and its closing value on the last trading day of the week.¹² The skipping of the nontrading period in the computation of returns follows Chordia and Subrahmanyam (2004) and is intended to mitigate the influence of bid-ask bounce on the relation between order imbalances and subsequent returns. While we use the visible S&P 500 index as our market proxy, our results are similar if we use alternative indices and index instruments, namely, the Russell 2000 and the Nasdaq-100 indices, or the S&P 500 ETF (SPY) and the S&P 500 E-mini futures, which is not surprising, as all returns on the indices and instruments are highly correlated with the return on the S&P 500 index.
- 6) *The CBOE Volatility Index (VIX)*. The VIX represents risk-neutral market expectation of volatility contained in prices of S&P 500 index options with approximately 30 days to expiration.

¹² Using excess returns rather than raw returns, as well as using S&P 500 index returns that incorporate the weekend return, and include dividends, produces essentially identical results.

7) *Macroeconomic indicators.* We also use three macroeconomic indicators as proxies for the business cycle: the short-term interest rate, the term spread and the credit spread (viz. Fama and French, 1989). The three-month constant maturity U.S. Treasury bill yield represents the short-term rate. The term spread is the difference between the 10-year and three-month constant maturity Treasury yields. The credit spread is the difference between the Moody's BAA and AAA yields. All yields are obtained from the FRED database of the Federal Reserve Bank of St. Louis. Each of the variables is computed by averaging the corresponding daily values within each week in the sample. Since the unit root null for the weekly time series of the T-bill yield, the term spread and the credit spread cannot be rejected in our sample, we use first differences for these variables.

Panel A of Table 1 presents the summary statistics for the above variables. Over our sample period, the mean (median) order imbalance for calls is -2.78% (-2.65%) and for puts the corresponding number is -1.48% (-1.43%). Thus, on average there are more customer sells of calls and puts. This contrasts with the view that dealers sell (or write) options to customers; indeed, over our sample customers are net sellers of options to dealers. The negative average order imbalances for ISE index options suggest that their clientele differs from that of the SPX options, which are characterized by positive average order imbalances (Gârleanu, Pedersen and Poteshman, 2009). The mean (median) OIB for NYSE stocks is also negative at -0.21% (-0.38%). On the other hand, the mean (median) OIB for the ETF SPY is 0.53% (0.49%). All the OIB variables show some evidence of positive skewness. The option OIBs are also more volatile than the OIBs for stocks, E-mini futures and the S&P 500 index ETF, SPY. This observation is confirmed by the OIB plots shown in Figure 1.

Panel B of Table 1 shows the summary statistics for the ISE index option trading volume. The average weekly trading volume is about 35,440 contracts for index calls and higher for index puts, at about 43,270 contracts. RUT options have higher average trading volumes than NDX options. Panel C of Table 1 reports the unconditional correlations. The correlation between order flows and index returns is especially strong for the index futures order imbalance but put option order flow is insignificantly related to contemporaneous index returns. OIB NYSE is positively correlated with OIB Calls and OIB E-mini. The VIX is negatively correlated with OIB E-mini, OIB NYSE, the index returns and the T-bill yield. The strong positive correlation between the VIX and the credit spread suggests that a high VIX points to worsening macroeconomic conditions.

3. Results

3.1. OLS Regression Results

To examine the information content of option order imbalances, we begin by estimating the following predictive regressions:

$$X_t = \alpha + \sum_{j=1}^3 \delta_j R_{t-j} + \sum_{j=1}^3 \beta_{1j} OIB\ Calls_{t-j} + \sum_{j=1}^3 \beta_{2j} OIB\ Puts_{t-j} + \sum_{j=1}^3 \gamma_j VIX_{t-j} + \varepsilon_t, \quad (1)$$

where X_t is the dependent variable (in turn, the S&P 500 return, the short-term rate, the term or credit spread, and the VIX), R_t is the S&P 500 return, $OIB\ Calls_t$ is order imbalance for index call options, and $OIB\ Puts_t$ is order imbalance for index put options.¹³ The idea that options trading may impact future returns is motivated by Back (1993), Biais and Hillion (1994), and Cao (1999), who argue that informed agents should be able to trade more profitably in the options

¹³ We include three lags in equation (1) for parsimony. In the next subsection, we use the Akaike criterion to select the number of lags, and formally confirm that three is the optimal number of lags.

market. Moreover, Easley, O'Hara and Srinivas (1998), Chakravarty, Gulen and Mayhew (2004) and Pan and Poteshman (2006) find that options trading contains information about future price movements. Lagged values of the VIX are included in the regression to capture possible delayed response of option order imbalances to the VIX.

The estimation results reported in Table 2 (with Newey-West corrected standard errors) provide evidence that the put option order imbalance is a useful predictor of aggregate stock returns and the VIX, whereas the call option OIB does not predict any of these variables. In particular, more position-opening put option buys relative to sells in the current week lead to an increase in the S&P 500 returns over the next two weeks. The two-week-ahead VIX also decreases in response to an increase in OIB Puts, pointing to improving market conditions. These estimates are economically significant. For example, a one-standard-deviation increase in OIB Puts predicts a 0.22% increase in the S&P 500 index return in the following week. While we would expect that bearish options order flow (buying of puts) should negatively predict returns if informed outside investors trade options, we find the opposite result, viz, a bearish order flow positively predicts returns.¹⁴

To provide a visual representation of the predictive relation between the put option order imbalance and aggregate stock returns, we sort the weekly OIB Puts into quintiles and compute the average S&P 500 index return over the next week and next two weeks for each quintile. Consistent with Table 2, Figure 2 shows a positive relation between OIB Puts and stock returns in subsequent weeks. Note that the returns generally increase as we move from the lowest to highest OIB quintiles, indicating that the results are not driven by extreme observations. Also, the first

¹⁴ The Internet Appendix considers out-of-sample tests, and shows that options order flow meaningfully forecasts index returns using, in turn, the years 2010, 2011, 2012, 2013, and 2014 as starting points for the out-of-sample forecasting period.

and last quintile portfolios earn similar two-week cumulative returns, indicating that our results do not arise exclusively from either net put buying or selling. A formal test indicates that the difference in average returns across the extreme quintiles is significantly different from zero over the next week as well as the next two weeks with p -values of less than 1%.

3.2. Vector Autoregression

Trading in the options market could predict returns and macroeconomic variables if options order flow reflects informed trading. However, past returns and macroeconomic conditions could also influence the hedging (or speculative) needs of investors and thus impact options trading. Given that there are reasons to expect bi-directional causality, we estimate the following vector autoregression (VAR) using weekly data:

$$\mathbf{x}_t = \boldsymbol{\alpha} + \sum_{j=1}^K \boldsymbol{\beta}_j \mathbf{x}_{t-j} + \boldsymbol{\varepsilon}_t, \quad (2)$$

where $\boldsymbol{\alpha}$ is a vector of constant terms, $\boldsymbol{\beta}_j$ is the vector of coefficients for lag j , $\mathbf{x}_t$ is a vector of the seven variables mentioned in the previous subsection (call option order imbalance, put option order imbalance, S&P 500 return, short-term rate, term spread, credit spread, and the VIX), and $\boldsymbol{\varepsilon}_t$ is a vector of random disturbances. The VAR uses three lags, i.e., $K = 3$, selected based on the Akaike information criterion.

Panel A of Table 3 presents the sum of the VAR coefficient estimates for the lagged option order imbalances. Chaboud, Chiquoine, Hjalmarsson and Vega (2014) argue that it is useful to conduct a test on the sum of the coefficients of the lags of the right-hand-side variable to measure both the magnitude and the direction of the long-run effect of the variable on the left-hand-side variable. The results indicate that a positive put option order imbalance predicts higher stock

returns. Thus, more position-opening put buys and/or fewer position-opening put sells causes the future S&P 500 returns to rise.¹⁵

Figure 3 plots cumulative impulse response functions (IRF) that represent the effect of a one standard-deviation innovation in option order imbalances on the S&P 500 returns for up to 10 weeks ahead. Using the notion that options trading activity should lead that in other contingent claims (Roll, Schwartz, and Subrahmanyam, 2014), and stock markets should lead other macroeconomic indicators, we use the following ordering for IRFs: OIB Calls, OIB Puts, S&P 500 Return, T-bill Yield, Term Spread, Credit Spread, VIX.¹⁶ The impulse responses suggest that there is a significant impact of the put order imbalance on future S&P 500 returns.¹⁷ The magnitudes of the impulse responses are economically meaningful. For example, a one-standard-deviation shock to the put order imbalance results in a cumulative S&P 500 return of about 0.55% over the next 10 weeks. Thus, an increase in the put order imbalance leads to an improvement in market conditions. These results are robust to an alternative ordering of the IRF when the S&P 500 return is first in the ordering sequence. Since the order imbalance is measured as the difference between the position-opening buy and sell orders, investors' purchases of puts from liquidity providers, alternatively market makers' sales of puts to outside investors, portend an increase in the S&P 500 returns.

Panel B of Table 3 presents the Wald test statistics for Granger causality tests. Consistent with the OLS results in Table 2, there is strong evidence that put order imbalances forecast index

¹⁵ Our finding that index option order flows contain information about future market returns is different from the conclusion of Muravyev, Pearson, and Broussard (2013) that no significant price discovery occurs in individual stock options. However, their results are based on analysis of tick-by-tick price adjustment, whereas we examine the ability of option order imbalances to predict weekly returns.

¹⁶ Generalized (order invariant) impulse responses are very similar to those shown in Figure 3.

¹⁷ To ensure that our results are not driven by the financial crisis, we estimate the same VAR in the sample period from January 2010 to December 2017 (the post-crisis period). The results are essentially similar to the ones reported in the paper and are available upon request.

returns. There is, however, no evidence that the imbalance Granger-causes macroeconomic indicators. Panel C reports the correlation matrix of the VAR residuals. Index return residuals are positively correlated with term spread residuals, and negatively correlated with credit spread and VIX residuals, suggesting that high index returns coincide with improved macroeconomic conditions as proxied by higher term and lower credit spreads, and lower VIX. Also, macroeconomic shocks are cross-correlated, as evidenced by the positive correlation between the VIX and the credit spread and the negative correlation between the VIX and the term spread. We also find that innovations to net put buying are positively related to VIX innovations and negatively related to index returns; we return to these findings in Section 5.

In untabulated results, we control for investor sentiment by including weekly sentiment measures based on the weekly sentiment survey of the American Association of Individual Investors, and the Investor Intelligence survey of investment advisors, as well as the Baker and Wurgler (2006) sentiment measure (which is only available at the monthly level). This is motivated by the work of Han (2008) who shows that options market activity is influenced by sentiment. We also include the economic policy uncertainty index of Baker, Bloom, and Davis (2016). These variables do not play a role in explaining S&P 500 index returns, and their inclusion leaves our other results unchanged.

3.3. How the Information Content of Option Order Imbalances Varies with Market Conditions

It is possible that the predictive ability of the put option order imbalances depends on market conditions. For example, standard informed trading models such as Kyle (1985) suggest that information has greater value when uncertainty is high. Also, relatively unsophisticated traders are more likely to trade put options to hedge their equity portfolios in periods of low stock returns

and high volatility, providing greater opportunities for other agents to trade on their private information. To account for this possibility, we estimate the following VAR:

$$\begin{aligned}\tilde{\mathbf{x}}_t = & \alpha + \sum_{j=1}^3 \beta_j \mathbf{x}_{t-j} + \sum_{j=1}^3 \theta_{1j} OIB\ Calls_{t-j} D_{t-j} + \sum_{j=1}^3 \theta_{2j} OIB\ Puts_{t-j} D_{t-j} \\ & + \sum_{j=1}^3 \gamma_j D_{t-j} + \varepsilon_t,\end{aligned}\tag{3}$$

where $\mathbf{x}_t$ is a vector that includes the call option order imbalance, put option order imbalance, S&P 500 return, short-term rate, term spread, and credit spread. In addition to these variables, $\tilde{\mathbf{x}}_t$ includes interaction terms between the call and put option order imbalances and a dummy variable D_t that captures stressful market conditions. This dummy variable is equal to one in weeks when the VIX is in its top quartile and zero otherwise. The model also includes three lags of the high-VIX dummy as exogenous variables.¹⁸

Panel A of Table 4 reports the sums of the coefficients of lagged call and put option order imbalances in the VAR with interaction terms shown in equation (3). There is strong evidence that the put option order imbalances predict returns only in periods of high perceived uncertainty as characterized by high VIX. In such periods, the magnitude of the long-run effect of put option order imbalances on returns increases by more than a factor of 10, from 0.02 to 0.29. Also, the put order imbalance leads to an increase in the term spread during the high VIX periods.

The cumulative impulse responses for this VAR specification are shown in Figure 4 and Panel B of Table 4. The following Cholesky ordering is used for the IRFs: OIB Calls, OIB Calls*High-VIX Dummy, OIB Puts, OIB Puts*High-VIX Dummy, S&P 500 Return, T-bill Yield, Term Spread, Credit Spread. An increase in put order imbalance results in at least a 10-week

¹⁸ The VIX is not included as an endogenous variable in this VAR specification because the model includes a dummy variable based on the level of the VIX.

impact on the S&P 500 return, both during and outside of the high VIX periods. There is some evidence that the order imbalance in calls has an impact on market returns during high VIX periods. However, this effect is not robust to changing the ordering of the IRFs (when the S&P 500 return is first in the ordering sequence); though the impact of OIB Puts remains robust.

4. Robustness Tests

For robustness we first interact OIB Puts, in turn, with two other variables associated with market stress, specifically, the TED spread (a proxy for financing costs) and a dummy for extreme down markets, i.e., whether the realized index return in a week is less than its full sample mean minus half its standard deviation (viz. Brunnermeier, 2009; Hameed, Kang, and Viswanathan, 2010). Including these variables does not alter the significance of the interaction of OIB Puts with the high VIX dummy.

In another robustness check, we compute the call-put implied volatility spread for the two index options included in our analysis (RUT and NDX) using the approach of Cremers and Weinbaum (2010) and add it to the VAR. Consistent with Atilgan, Bali and Demirtas (2015), this variable is a significant predictor of the S&P 500 return in the next week. The implied volatility spread is uncorrelated with option order imbalances and adding it to the VAR has essentially no effect on the coefficients of the put order imbalance.

Bollerslev, Tauchen and Zhou (2009) show that the difference between implied and realized variance (termed variance risk premium or VRP) positively predicts the overall stock market returns, and Han and Zhou (2011) show that it predicts returns in the cross-section. When this variable is added to the VAR model in equation (2), it is a positive and significant predictor

of the S&P 500 returns. However, the predictive ability of OIB Puts remains unchanged. Similarly, adding the CBOE Skew (tail risk) measure to the VAR makes no material difference to the results.

The VAR in equation (3) uses a dummy variable for high-volatility periods that is based on the level of the VIX. This model assumes that shifts between low- and high-volatility periods are deterministic events. An alternative way to examine state dependence in the predictive ability of OIB Puts is to use a Markov-switching regression model. An attractive characteristic of regime-switching models is that regime probabilities are treated as unknown parameters to be estimated together with the model coefficients. Such models capture the intuition that, although economic and market conditions evolve randomly, investors can make probabilistic inferences about the underlying state using observable data. We estimate the following Markov-switching specification for the S&P 500 index returns:

$$R_t = \alpha + \sum_{j=1}^3 \theta_j R_{t-j} + \sum_{j=1}^3 \beta_j OIB\ Calls_{t-j} + \sum_{j=1}^3 \gamma_j VIX_{t-j} + \sum_{j=1}^3 \delta_{j,s_t} OIB\ Puts_{t-j} + \varepsilon_t, \quad (4)$$

where $\varepsilon_t \sim N(0, \sigma_{s_t}^2)$ and the unobserved state variable $s_t = \{1, 2\}$ follows a Markov process with fixed transition probabilities:¹⁹

$$p_{11} = P(s_t = 1 | s_{t-1} = 1),$$

$$p_{22} = P(s_t = 2 | s_{t-1} = 2).$$

Since the error variance is assumed to be state-dependent, the model allows the coefficients of OIB Puts to vary between the low- and high-variance states. A brief technical background on Markov-switching models is provided in Appendix A.

The estimation results are shown in Table 5. The estimated standard deviation of the model errors is about 2.7 times as high in state 2 as in state 1. Therefore, state 1 (2) can be

¹⁹ A model with transition probabilities dependent on the previous value of the VIX produces similar results.

interpreted as a low- (high-) variance state. Based on the estimated transition probabilities, the expected durations of these states are about 52 weeks and 19 weeks, respectively. The put option order imbalance has predictive power for future stock returns only in the high-variance state. This result is consistent with the VAR results in Panel A of Table 4. Figure 5 shows the filtered probability of the high-variance state. The correlations of this estimated probability with the VIX and the high-VIX dummy are about 0.78 and 0.82, respectively.

It is possible that option order flow proxies for order flow from other contingent claims on the S&P 500. Accordingly, we account for non-option order imbalances, including those for the E-mini S&P 500 futures, the underlying index, and the SPY ETF. These order imbalances are added to the VAR in equation (3) as additional endogenous variables.

Panel A of Table 6 reports the results of the Wald test on the sums of the coefficients of lagged order imbalances in the VAR. Non-option order imbalances are not significant predictors of stock returns. The results for option order imbalances are very similar to the results in Table 4: put option imbalances positively predict stock returns in periods characterized by high levels of the VIX. The fact that these results hold after controlling for non-option order imbalances suggests that put option trades contain some information that is unique to the option market.

Cumulative impulse responses for the VAR with non-option order imbalances are shown in Panel B of Table 6. There is evidence that the E-mini order imbalance has a significant long-run effect on stock returns. The IRF plots in the Internet Appendix show that this effect is contemporaneous, whereas the put option order imbalance predicts returns in subsequent weeks but has no effect on contemporaneous returns. The correlation between the VAR residuals for the E-mini order imbalance and the S&P 500 return is about 0.60. The strong contemporaneous relation between the E-mini order imbalance and stock returns is consistent with prior work such

as Evans and Lyons (2002) and Brandt and Kavajecz (2004) for currency and Treasury markets, respectively.

The ISE data on index options, which we extensively use here, do not include the S&P 500 index options (SPX) because they are traded only at the CBOE, which has an exclusive agreement with S&P and Dow Jones Indices. Because SPX options are some of the most actively traded options, it is important to confirm the robustness of our main results by controlling for SPX option order imbalances. To accomplish this, we obtain the open-close data for SPX options from Market Data Express, an exclusive provider of CBOE historical data. These open-close data have exactly the same structure and methodology as the ISE data, as the Options Clearing Corporation requires homogeneous reporting across option exchanges. Thus, we compute SPX option order imbalances in the same exact way as for the ISE index options to ensure a fair comparison.

In Table 7, we test whether controlling for SPX option order flow affects the ability of ISE put order flow to predict market returns. We tried a number of different specifications that produced consistent results but given limited space only report IRFs in Table 7. The magnitude of the ISE order flow predictability remains unchanged when SPX option order imbalances are added to the VAR. There is also no evidence that SPX option order flow predicts index returns. Thus, the predictive ability of put option order flow at the weekly horizon appears special to ISE options. As we explain below, SPX options and ISE index options differ in several important ways.

The analysis of order flows from other index derivatives suggests that the hedging of speculative positions in other securities using options markets is not the primary driver of our results. Specifically, the fact that index return predictability from ISE index put options survives controls for order flow from other index derivatives and the spot market suggests that agents are

not using these other instruments to speculate on information. If this were true, order flow from these other instruments would interfere with return predictability from index put options.

5. Alternative Explanations for our Central Result

Thus far, we have documented that put option order imbalance predicts weekly index returns, and this predictive ability supersedes that from other order flows. In Section 3.2, we considered and ruled out investor sentiment as potential explanations for our finding. In this section, we consider additional explanations for this phenomenon.

5.1. Is the Return Predictability Due to Informed Trading by Market Makers?

The contrarian nature of our finding is consistent with the hypothesis that market makers (who take the other side of customers' orders) find it profitable to trade on private information in the put options markets. If the predictive ability of put option order imbalances is indeed explained by informed trading, this predictive power should become stronger around important information events, such as major macroeconomic announcements. To test this hypothesis, we estimate the following VAR in which call and put option order imbalances are interacted with a dummy for major macroeconomic announcements:

$$\begin{aligned} \tilde{\mathbf{x}}_t = & \alpha + \sum_{j=1}^3 \beta_j \mathbf{x}_{t-j} + \sum_{j=1}^3 \theta_{1j} OIB\ Calls_{t-j} (1 - D_{t-j}) + \sum_{j=1}^3 \theta_{2j} OIB\ Calls_{t-j} D_{t-j} \\ & + \sum_{j=1}^3 \theta_{3j} OIB\ Puts_{t-j} (1 - D_{t-j}) + \sum_{j=1}^3 \theta_{4j} OIB\ Puts_{t-j} D_{t-j} + \sum_{j=1}^3 \gamma_j D_{t-j} + \varepsilon_t, \end{aligned} \quad (5)$$

where $\mathbf{x}_t$ is a vector of five variables, namely, the S&P 500 return, the short rate, the term and credit spreads, and the VIX. In addition to these variables, vector $\tilde{\mathbf{x}}_t$ contains four interaction variables between option order imbalances and a dummy variable D_t . This dummy is equal to one

in weeks that contain the U.S. Consumer Price Index (CPI), Gross Domestic Product (GDP), unemployment announcements, and Federal Open Market Committee (FOMC) meetings, and zero otherwise. We follow Fleming and Remolona (1999) in choosing the first three announcements and rely on Piazzesi (2005) to motivate the FOMC meetings. The model also includes three lags of the news dummy as exogenous variables.

Panel A of Table 8 shows the sums of the VAR coefficient estimates for the option OIBs. We find that put order flow is a significant predictor of stock returns only in weeks containing the four macroeconomic announcements. But note that put order flow differentially predicts returns *after* the announcements. Indeed, as shown in Panel B of Table 8, we do not find any evidence of differential return predictability around macroeconomic announcements when the news dummy is defined as being one in the week preceding the announcement and zero otherwise. Indeed, the point estimates of the coefficients are very similar for the week preceding the announcement and other weeks. This does not accord with intense informed trading prior to the announcements.

If market makers exploit information by changing bid and ask quotes strategically, then they would move bid and ask prices up to encourage sells and discourage buys when they have positive information. Subsequently when market maker information is incorporated into prices, put returns should be positive. A symmetric argument holds for the case when market maker information is negative. This argument implies high midquote returns should be followed by high returns. In fact, weekly midquote returns in puts are weakly *negatively* autocorrelated. Specifically, the mean delta-adjusted serial correlation in individual put weekly midquote returns is -10%. Further, put midquote returns do not significantly predict the subsequent weeks' put order flow, indicating that market makers' quote setting strategies do not attract order flow in the desired

direction to facilitate informed trading. These results suggest that informed trading by market makers is not the primary cause of our results.

5.2. *The Demand for Insurance*

Another related explanation for our results is that OIB Puts is driven by investors' demand for insurance. Market makers offer this insurance by selling index puts. In times of market stress, the demand for insurance increases. The expected market risk premium increases as well, leading to a positive relation between public put buying and subsequent market excess returns. On the other hand, in good times, people sell puts in the hope of earning the put premium. This hypothesis is consistent with Grossman and Zhou (1996) and Bates (2008). In their models, less crash-averse agents (the intermediary firms making markets) insure the more crash-averse agents (outside investors) through options.

Supporting the insurance explanation, Panel C of Table 3 shows that there is indeed a positive relation between innovations to OIB Puts and the VIX (correlations of 0.11). We also observe that innovations to OIB Puts are negatively correlated with innovations to index returns (correlation of -0.09). These results accord with the view that investors buy puts during periods of high uncertainty and in down-markets.

We would expect put buying during uncertain periods to be stronger than put writing during periods with low uncertainty because one could achieve greater upside in the latter case through alternatives such as buying other derivatives. To further investigate the above explanation, we divide the option order imbalance into positive and negative order imbalances. Specifically we define positive OIB as $\max(OIB, 0)$ and negative OIB as $\min(OIB, 0)$. The corresponding VAR results are reported in Table 9. We find that positive OIB puts are related to an increase in S&P

500 returns while negative OIB Puts are not related to future returns. The evidence from the previous section that OIB Puts predicts returns following macroeconomic announcements also suggests that divergence of opinion following the announcement (Kim and Verrecchia, 1994) increases uncertainty to which investors respond by buying puts. Our result that put option order flow predicts returns more strongly during high VIX periods, coupled with the result in Table 9, further supports the notion that increased put buying during uncertain times, which raises required market returns, is the driver of return predictability from put order flow. From Section 3.2, the economic magnitude of the predictability, about a 2.9% annualized market return for a one standard deviation move in options order flow, is reasonable enough to be consistent with a risk-based explanation.²⁰

In Section 4, we find that order flows in SPX options do not predict returns. This raises the puzzle of why ISE options would be suited for insurance but not SPX options. We believe the answer lies in the notion that buying puts to reduce downside risk is more desirable for agents with greater risk aversion, and retail investors are likely to be more risk averse than larger institutional investors (Haddad and Muir, 2018). In two untabulated tests, we confirm that order flow for ISE index options tends to consist of retail orders to a larger extent than for S&P 500 index options. Institutional investors prefer SPX options because of the ability to execute large multi-leg trades and relatively low transaction costs. Other index options, including the most liquid ISE index options that we study, are traded by few investors with high preference for a given index and mostly by unsophisticated retail investors. First, using intraday options transaction data from OPRA from 2006 to 2015, we find that the average dollar trade size for index options on the ISE

²⁰ For comparison, over the 1926-2019 period, the excess market return, and the size and value premia, as defined in Fama and French (1998), are about 7.9%, 2.5%, and 4.4% per year, respectively (from the data on the Fama-French factors at the Wharton Research Data Services (WRDS) website).

is one-sixth of that for SPX options. Second, we note that the open-close data label trading volume as “firm” when a member like Morgan Stanley trades for their own account, while other non-market-maker trades that include retail trades are marked as “customer.” Thus, firm volume comes entirely from sophisticated institutions, while retail investor trades are part of customer volume. For position-opening trades, we compute the fraction of total volume that corresponds to customer volume. Consistent with the hypothesis that most order flow in ISE index options is retail, customer trades constitute about 90% of position-opening volume for ISE index options, which is substantially larger than the corresponding fraction for CBOE (69%). These tests support the notion that ISE customers tend to be smaller retail investors.

The ISE Open/Close Trade Profile data subdivides volumes of customer trades into those of small trades (not exceeding 100 contracts per trade), medium trades (ranging from 101 to 200 contracts per trade) and large trades (larger than 200 contracts per trade). We decompose the option order imbalances into two parts: OIBs of small customer trades and OIBs of larger customer trades and firm trades. These order imbalances are computed by dividing the difference between the corresponding buy and sell volumes by the total volume of customer and firm trades. In the Internet Appendix, we show the impulse responses for the VAR containing the two components of the call and put option order imbalances. Based on these results, only order imbalances of small customer put trades contain information about future S&P 500 index returns. This finding supports the argument above that retail customers use puts to hedge risk exposures.

5.3. Why Does Index Return Predictability Emanate from Order Flow in Puts but not Calls?

We note that for the overall sample, put order flow predicts returns far more strongly than does the order flow for calls. Specifically, from Table 9, we find that neither net call buying nor net call

selling predict future returns. This is an interesting puzzle. One possible explanation for call order flow not predicting returns is that puts provide greater protection on the downside than call writing, because with negative returns the call writers only collect the premiums but put buyers capture the negative returns. Thus, puts may attract more retail orders than calls. Indeed, in untabulated results using intraday transaction data, we find that on the ISE, the average dollar trade size is \$22,419 for index puts but \$27,720 for index calls, a 24% differential.

In Table 10, we provide basic statistics for bid-ask spreads and trading volume for ISE index options. Panels A and B of the Table show that index puts are significantly more liquid than index calls on average.²¹ The difference between the liquidity of call and put options increases in periods of high VIX because the liquidity of index puts increases substantially in such periods. Panel C of Table 10 compares the average bid-ask spreads for call and put options during weeks with and without major U.S. macroeconomic announcements (FOMC, CPI, GDP, and unemployment). This univariate analysis shows that the bid-ask spreads of index puts (but not calls) are significantly lower on average during weeks with major macroeconomic announcements than in other weeks. Bollen and Whaley (2004) also indicate that index puts are much more actively traded than calls. All this evidence is consistent with the popularity of puts for insurance purposes.

6. Conclusion

Given the likelihood of informed agents preferring options owing to greater leverage, it is reasonable that individual stock options might play an informational role. It is less clear whether

²¹ The data used to compute option bid-ask spreads and volume are obtained from OptionMetrics. While we use averages of volume-weighted spreads in Table 10 to avoid giving undue weight to infrequently-traded options, the results are similar if simple averages are used.

index options might play such a role, since private information about the overall market is harder to come by. We find that net buying pressure in ISE index put options positively predicts S&P 500 index returns. This result obtains even though index order flow does not predict index returns.

We explore a number of possible rationales for our result. Our results are most consistent with the notion that investors buy protection using put options when uncertainty is high. The uncertainty is accompanied by higher required market returns, thus yielding our predictability result. This is supported by the finding that the predictability is strongest around periods with scheduled macroeconomic news announcements (when uncertainty is high). Further, put option bid-ask spreads are lower, whereas the call option spreads are the same, in weeks with major macroeconomic announcements as compared to weeks without the major macroeconomic announcements, suggesting that increased put demand during periods of high uncertainty adds to liquidity in put options markets.

Our analysis suggests several areas for future exploration. For example, whether our results also hold in international settings would be of interest. The basic notion is that the more the uncertainty in the relevant country, the more likely are index options to play an informational role. It would be of interest to further decompose the outside investor clientele of index options investors and ascertain whether the sophistication of this clientele influences the demand for put-based insurance. These and other topics are left for future research.

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Table 1
Summary Statistics for Order Imbalances, S&P 500 Returns and Macroeconomic Indicators

Panel A reports the summary statistics for variables used in the analysis. In addition to the full sample period, the statistics are computed separately for the weeks when the VIX is in its top quartile and for the rest of the sample. Panel B shows the summary statistics for the index option trading volume, expressed in thousands of contracts. Panel C presents the Pearson correlations for variables used in the analysis. Bold text indicates statistical significance at 5% level and *p*-values are shown in parentheses. All variables are measured at weekly intervals. All variables in Panels A and C except the T-bill yield, the term spread and the credit spread are expressed in percentage terms. First differences (in basis points) are used for the T-bill yield, the term spread and the credit spread. The sample period is from January 2, 2006 to December 29, 2017 and contains 615 observations.

Panel A: Summary Statistics

	OIB Calls	OIB Puts	OIB E-mini	OIB NYSE	OIB SPY	S&P 500 Return	T-bill Yield	Term Spread	Credit Spread	VIX
Full sample (N=615)										
Mean	-2.78	-1.48	0.01	-0.21	0.53	0.11	-0.45	0.15	-0.04	19.27
Median	-2.65	-1.43	-0.03	-0.38	0.49	0.20	0.05	-0.80	0.00	16.43
Std. Dev.	7.16	5.93	1.48	3.07	3.09	2.42	9.36	12.59	4.96	9.64
Skewness	0.12	0.08	0.20	0.52	0.52	-0.97	-3.55	0.56	2.04	2.48
Kurtosis	4.93	4.71	3.15	4.29	5.88	12.72	42.58	13.29	25.20	11.43
VIX below top quartile (N=459)										
Mean	-2.47	-1.70	0.15	0.14	0.61	0.41	0.58	0.19	-0.29	15.00
Median	-2.41	-1.58	0.13	-0.05	0.56	0.36	0.20	-0.45	-0.20	14.38
Std. Dev.	7.45	6.00	1.48	3.28	3.36	1.45	4.68	8.69	2.72	3.26
Skewness	0.07	0.07	0.11	0.44	0.53	-0.03	-2.22	0.65	-0.40	0.46
Kurtosis	4.91	4.88	3.24	3.85	5.37	3.46	42.79	4.76	5.23	2.32
VIX in top quartile (N=156)										
Mean	-3.71	-0.82	-0.41	-1.26	0.33	-0.80	-3.49	0.00	0.68	31.82
Median	-3.58	-0.98	-0.57	-1.19	0.37	-0.84	-0.60	-1.90	0.60	27.49
Std. Dev.	6.18	5.71	1.38	2.06	2.08	3.98	16.43	20.12	8.65	11.13
Skewness	0.16	0.16	0.46	-0.57	-0.30	-0.26	-1.94	0.41	1.34	1.98
Kurtosis	4.47	4.11	3.21	3.82	3.84	6.15	15.42	7.63	10.19	6.99

Panel B: Summary Statistics for Position-Opening Option Trading Volumes

	Calls	Puts	NDX Calls	NDX Puts	RUT Calls	RUT Puts
Mean	35.44	43.27	10.35	14.81	40.40	45.82
Median	27.80	37.43	8.22	11.94	35.94	43.00
Std. Dev.	34.99	38.11	8.51	12.24	28.20	26.58
Skewness	1.38	1.23	1.22	1.84	1.46	1.38
Kurtosis	2.53	2.29	1.50	5.05	3.17	4.01
N	615	615	615	615	382	382

Panel C: Correlations of Order Imbalances (OIBs), Index Returns and Macroeconomic Indicators

	OIB Calls	OIB Puts	OIB E-mini	OIB NYSE	OIB SPY	S&P 500 Return	Nasdaq- 100 Return	Russell 2000 Return	T-bill Yield	Term Spread	Credit Spread
OIB Puts	0.11 (0.01)										
OIB E-mini	0.04 (0.31)	-0.01 (0.72)									
OIB NYSE	0.16 (0.00)	0.07 (0.10)	0.37 (0.00)								
OIB SPY	0.08 (0.04)	-0.06 (0.13)	0.21 (0.00)	0.16 (0.00)							
S&P 500 Return	0.09 (0.02)	-0.05 (0.19)	0.60 (0.00)	0.31 (0.00)	0.22 (0.00)						
Nasdaq-100 Return	0.08 (0.04)	-0.03 (0.49)	0.57 (0.00)	0.26 (0.00)	0.18 (0.00)	0.87 (0.00)					
Russell 2000 Return	0.10 (0.02)	-0.05 (0.22)	0.56 (0.00)	0.27 (0.00)	0.21 (0.00)	0.92 (0.00)	0.85 (0.00)				
T-bill Yield	-0.02 (0.56)	-0.02 (0.70)	0.05 (0.26)	0.07 (0.09)	-0.00 (0.91)	-0.01 (0.80)	-0.01 (0.90)	-0.05 (0.26)			
Term Spread	0.05 (0.22)	-0.05 (0.20)	0.12 (0.00)	0.05 (0.26)	0.05 (0.19)	0.26 (0.00)	0.22 (0.00)	0.27 (0.00)	-0.64 (0.00)		
Credit Spread	-0.01 (0.76)	0.08 (0.06)	-0.02 (0.66)	-0.05 (0.23)	0.08 (0.05)	-0.10 (0.01)	-0.13 (0.00)	-0.12 (0.00)	-0.04 (0.36)	-0.09 (0.03)	
VIX	-0.06 (0.11)	0.03 (0.46)	-0.17 (0.01)	-0.23 (0.00)	-0.03 (0.51)	-0.29 (0.00)	-0.24 (0.00)	-0.27 (0.00)	-0.15 (0.00)	-0.04 (0.28)	0.28 (0.00)

Table 2

Regressions of S&P 500 Returns and Macroeconomic Indicators on Lagged Order Imbalances

This table reports estimation results for the following predictive regressions:

$X_t = \alpha + \sum_{j=1}^3 \delta_j R_{t-j} + \sum_{j=1}^3 \beta_{1j} OIB\ Calls_{t-j} + \sum_{j=1}^3 \beta_{2j} OIB\ Puts_{t-j} + \sum_{j=1}^3 \gamma_j VIX_{t-j} + \varepsilon_t$, where X_t is the dependent variable (S&P 500 return, T-bill yield, term spread, credit spread or the VIX), $OIB\ Calls_t$ is order imbalance for call options, and $OIB\ Puts_t$ is order imbalance for put options. All variables are measured at weekly intervals. All variables except the T-bill yield, the term spread and the credit spread are expressed in percentage terms. First differences (in basis points) are used for the T-bill yield, the term spread and the credit spread. The sample period is from January 2, 2006 to December 29, 2017 and contains 615 observations. The regressions are estimated using OLS with the Newey-West heteroskedasticity and autocorrelation consistent covariance matrix. Bold text indicates statistical significance at 5% level.

	S&P 500 Return		T-bill Yield		Term Spread		Credit Spread		VIX	
	Coef.	t-stat	Coef.	t-stat	Coef.	t-stat	Coef.	t-stat	Coef.	t-stat
Intercept	0.3491	1.11	1.3143	1.35	0.1969	0.12	-0.3537	-0.36	0.9017	2.70
R_{t-1}	-0.0291	-0.34	0.2929	0.55	0.1817	0.25	-0.4913	-2.50	-0.0588	-0.50
R_{t-2}	0.0547	0.45	-0.1034	-0.22	0.3463	0.50	-0.5902	-2.04	-0.1124	-0.62
R_{t-3}	-0.0869	-1.27	0.3149	1.39	-0.4961	-1.61	-0.2732	-2.51	0.0846	1.08
$OIB\ Calls_{t-1}$	0.0036	0.33	-0.0034	-0.07	-0.1067	-1.80	-0.0228	-1.12	0.0011	0.07
$OIB\ Calls_{t-2}$	-0.0094	-0.82	0.0247	0.57	-0.0236	-0.40	0.0204	-0.88	-0.0002	-0.01
$OIB\ Calls_{t-3}$	-0.0092	-0.66	-0.0221	-0.44	0.0855	1.27	0.0286	1.17	-0.0006	-0.03
$OIB\ Puts_{t-1}$	0.0366	2.19	-0.0932	-1.33	0.0818	0.90	0.0219	0.73	-0.0342	-1.73
$OIB\ Puts_{t-2}$	0.0527	3.14	-0.0439	-0.89	0.1409	1.92	0.0143	0.54	-0.0528	-2.10
$OIB\ Puts_{t-3}$	0.0010	0.07	0.0609	0.85	-0.0646	-0.72	0.0474	1.55	-0.0075	-0.36
VIX_{t-1}	0.0245	0.41	-0.2136	-0.50	0.4690	1.02	0.2308	1.14	0.7668	7.78
VIX_{t-2}	-0.0241	-0.25	0.1401	0.40	-0.3469	-0.72	-0.0923	-0.67	0.1224	0.72
VIX_{t-3}	-0.0083	-0.09	-0.0282	-0.10	-0.1203	-0.35	-0.1105	-0.76	0.0579	0.43
Adjusted R^2	0.0238		0.0290		0.0115		0.2575		0.8822	

Table 3
Vector Autoregression Results

This table reports results from the following VAR:

$\mathbf{x}_t = \boldsymbol{\alpha} + \sum_{j=1}^3 \boldsymbol{\beta}_j \mathbf{x}_{t-j} + \boldsymbol{\varepsilon}_t$, where $\boldsymbol{\alpha}$ is a vector of constant terms, $\boldsymbol{\beta}_j$ is the vector of coefficients for lag j , $\mathbf{x}_t$ is a vector of seven variables (OIB Calls, OIB Puts, S&P 500 return, T-bill yield, term spread, credit spread, and the VIX), and $\boldsymbol{\varepsilon}_t$ is a vector of random disturbances. All variables are measured at weekly intervals. All variables except the T-bill yield, the term spread and the credit spread are expressed in percentage terms. First differences (in basis points) are used for the T-bill yield, the term spread and the credit spread. Panel A presents the sums of the VAR coefficients on the lags of OIBs. The reported Wald test statistics are for the tests that the corresponding sums of coefficients are equal to zero. Panel B presents the Wald test statistics of VAR Granger causality (block exogeneity) tests. The null hypothesis is that row variable does not forecast the column variable and the p -values are shown in parentheses. Panel C reports the Pearson correlations of the residual series with p -values in parentheses. The sample period is from January 2, 2006 to December 29, 2017 and contains 615 observations. Bold text indicates statistical significance at 5% level.

Panel A: Sums of VAR Coefficients for Option Order Imbalances (OIBs)

	S&P 500 Return	T-bill Yield	Term Spread	Credit Spread	VIX
Sum of coefficients on lags of S&P 500 return	0.0138	0.6105	-0.4235	-0.7568	-0.1326
Wald Chi-square statistic (Sum=0)	0.0144	1.9017	0.4964	18.8599	0.6881
p -value	0.9046	0.1679	0.4811	0.0000	0.4068
Sum of coefficients on lags of OIB Calls	-0.0145	0.0076	-0.0414	-0.0007	-0.0001
Wald Chi-square statistic (Sum=0)	0.4851	0.0089	0.1447	0.0005	0.00002
p -value	0.4861	0.9247	0.7037	0.9817	0.9961
Sum of coefficients on lags of OIB Puts	0.0956	-0.0828	0.2074	0.0178	-0.1050
Wald Chi-square statistic (Sum=0)	14.2573	0.7198	2.4490	0.2154	8.8693
p -value	0.0002	0.3962	0.1176	0.6425	0.0029

Panel B: Granger Causality Tests

	OIB Calls	OIB Puts	S&P 500 Return	T-bill Yield	Term Spread	Credit Spread	VIX
OIB Calls	—	5.55 (0.14)	0.96 (0.81)	0.44 (0.93)	3.41 (0.33)	2.83 (0.42)	0.05 (0.99)
OIB Puts	12.38 (0.01)	—	18.00 (0.00)	4.08 (0.25)	6.01 (0.11)	1.24 (0.74)	10.28 (0.02)
S&P 500 Return	14.82 (0.00)	1.59 (0.66)	—	6.30 (0.10)	5.23 (0.16)	25.23 (0.00)	2.57 (0.46)
T-bill Yield	1.56 (0.67)	12.64 (0.01)	6.39 (0.09)	—	4.84 (0.18)	11.38 (0.01)	6.35 (0.10)
Term Spread	1.07 (0.78)	5.36 (0.15)	6.00 (0.11)	5.80 (0.12)	—	3.12 (0.37)	4.74 (0.19)
Credit Spread	2.14 (0.54)	9.02 (0.03)	6.10 (0.11)	11.03 (0.01)	9.24 (0.03)	—	2.73 (0.44)
VIX	4.30 (0.23)	0.88 (0.83)	0.60 (0.90)	2.15 (0.54)	3.65 (0.30)	4.15 (0.25)	—
All	33.31 (0.02)	34.23 (0.01)	44.41 (0.00)	49.88 (0.00)	39.56 (0.00)	141.95 (0.00)	34.67 (0.00)

Panel C: Correlations of VAR Residuals

	OIB Calls	OIB Puts	S&P 500 Return	T-bill Yield	Term Spread	Credit Spread
OIB Puts	0.05 (0.22)					
S&P 500 Return	0.07 (0.10)	-0.09 (0.03)				
T-bill Yield	-0.02 (0.57)	0.00 (0.99)	0.01 (0.72)			
Term Spread	0.03 (0.47)	-0.07 (0.10)	0.24 (0.00)	-0.64 (0.00)		
Credit Spread	-0.03 (0.44)	0.04 (0.29)	-0.14 (0.00)	0.03 (0.51)	-0.11 (0.01)	
VIX	-0.02 (0.64)	0.11 (0.00)	-0.82 (0.00)	-0.06 (0.15)	-0.18 (0.00)	0.14 (0.00)

Table 4
Sums of VAR Coefficients and Impulse Response Functions
for VAR with interaction terms for high VIX periods

This table reports results from the following VAR:

$\tilde{\mathbf{x}}_t = \boldsymbol{\alpha} + \sum_{j=1}^3 \boldsymbol{\beta}_j \mathbf{x}_{t-j} + \sum_{j=1}^3 \boldsymbol{\theta}_{1j} OIB\ Calls_{t-j} D_{t-j} + \sum_{j=1}^3 \boldsymbol{\theta}_{2j} OIB\ Puts_{t-j} D_{t-j} + \sum_{j=1}^3 \boldsymbol{\gamma}_j D_{t-j} + \boldsymbol{\varepsilon}_t$, where $\mathbf{x}_t$ is a vector that includes the call option order imbalance, put option order imbalance, S&P 500 return, T-bill yield, term spread, and credit spread. In addition to these variables, $\tilde{\mathbf{x}}_t$ includes interaction terms between the call and put option order imbalances and a dummy variable D_t equal to one in weeks when the VIX is in its top quartile and zero otherwise. All variables are measured at weekly intervals. First differences (in basis points) are used for the T-bill yield, the term spread and the credit spread. Panel A shows the sums of VAR coefficients on lags of option order imbalances in the equations shown in column headings. The reported Wald test statistics are for the tests that the corresponding sums of coefficients are equal to zero. Panel B presents the accumulated responses of column variables to Cholesky one-standard-deviation innovations in row variables for a forecast horizon of 10 weeks. The following Cholesky ordering is used: OIB Calls, OIB Calls*High-VIX Dummy, OIB Puts, OIB Puts*High-VIX Dummy, S&P 500 Return, T-bill Yield, Term Spread, Credit Spread. Standard errors are shown in parentheses. The sample period is from January 2, 2006 to December 29, 2017 and contains 615 observations, including 156 observations for which the high-VIX dummy is equal to one. Bold text indicates statistical significance at 5% level.

Panel A: Sums of VAR Coefficients

	S&P 500 Return	T-bill Yield	Term Spread	Credit Spread
Sum of coefficients on lags of OIB Calls				
Estimate	-0.0089	0.1011	-0.0508	-0.0114
Wald Chi-square statistic (Sum=0)	0.1510	1.3157	0.1705	0.1020
<i>p</i> -value	0.6975	0.2514	0.6797	0.7495
Sum of coefficients on lags of interaction of OIB Calls with high-VIX dummy				
Estimate	0.0345	-0.7035	0.2927	0.0315
Wald Chi-square statistic (Sum=0)	0.4214	11.8728	1.0562	0.1442
<i>p</i> -value	0.5163	0.0006	0.3041	0.7042
Sum of coefficients on lags of OIB Calls and interaction of OIB Calls with high-VIX dummy				
Estimate	0.0256	-0.6024	0.2419	0.0201
Wald Chi-square statistic (Sum=0)	0.2928	11.0040	0.9121	0.0739
<i>p</i> -value	0.5884	0.0009	0.3396	0.7857
Sum of coefficients on lags of OIB Puts				
Estimate	0.0235	-0.0620	0.1010	0.0104
Wald Chi-square statistic (Sum=0)	0.6730	0.3168	0.4313	0.0544
<i>p</i> -value	0.4120	0.5735	0.5113	0.8156
Sum of coefficients on lags of interaction of OIB Puts with high-VIX dummy				
Estimate	0.2635	-0.0364	0.4153	0.0769
Wald Chi-square statistic (Sum=0)	21.9040	0.0282	1.8924	0.7637
<i>p</i> -value	0.0000	0.8666	0.1689	0.3822
Sum of coefficients on lags of OIB Puts and interaction of OIB Puts with high-VIX dummy				
Estimate	0.2870	-0.0984	0.5163	0.0873
Wald Chi-square statistic (Sum=0)	34.6798	0.2758	3.9024	1.3149
<i>p</i> -value	0.0000	0.5994	0.0482	0.2515

Panel B: Impulse Responses

	S&P 500 Return	T-bill Yield	Term Spread	Credit Spread
OIB Calls	0.15 (0.28)	-0.49 (0.97)	0.55 (1.37)	-0.47 (0.98)
OIB Calls*High-VIX Dummy	0.65 (0.29)	-2.69 (1.01)	2.55 (1.43)	-0.14 (1.03)
OIB Puts	0.55 (0.27)	-0.46 (0.94)	0.55 (1.33)	-0.01 (0.96)
OIB Puts*High-VIX Dummy	0.83 (0.26)	0.80 (0.91)	-0.37 (1.29)	-0.57 (0.94)

Table 5
Markov-Switching Model of S&P 500 Returns

This table shows parameter estimates for the following Markov-switching model:

$R_t = \alpha + \sum_{j=1}^3 \theta_j R_{t-j} + \sum_{j=1}^3 \beta_j OIB\ Calls_{t-j} + \sum_{j=1}^3 \gamma_j VIX_{t-j} + \sum_{j=1}^3 \delta_{j,s_t} OIB\ Puts_{t-j} + \varepsilon_t$, where R_t is the S&P 500 index return, $\varepsilon_t \sim N(0, \sigma_{\varepsilon_t}^2)$, and the unobserved state variable $s_t = \{1, 2\}$ follows a Markov process with fixed transition probabilities: $p_{11} = P(s_t = 1 | s_{t-1} = 1)$, $p_{22} = P(s_t = 2 | s_{t-1} = 2)$. All variables are measured at weekly intervals. The sample period is from January 2, 2006 to December 29, 2017 and contains 615 observations. The z-statistics are based on Huber-White robust standard errors. Bold text indicates statistical significance at 5% level.

		Estimate	z-statistic	p-value
State 1 (low-variance)	<i>OIB Puts</i> _{<i>t</i>-1}	0.0035	0.26	0.7983
	<i>OIB Puts</i> _{<i>t</i>-2}	0.0092	0.68	0.4953
	<i>OIB Puts</i> _{<i>t</i>-3}	0.0026	0.18	0.8599
	σ_1	1.3825	3.81	0.0001
State 2 (high-variance)	<i>OIB Puts</i> _{<i>t</i>-1}	0.1348	2.23	0.0258
	<i>OIB Puts</i> _{<i>t</i>-2}	0.2081	2.85	0.0043
	<i>OIB Puts</i> _{<i>t</i>-3}	0.0175	0.28	0.7824
	σ_2	3.7434	7.45	0.0000
Common regressors	Intercept	-0.1444	-0.59	0.5529
	<i>R</i> _{<i>t</i>-1}	-0.0713	-0.97	0.4238
	<i>R</i> _{<i>t</i>-2}	0.0698	0.95	0.3445
	<i>R</i> _{<i>t</i>-3}	-0.0607	-1.31	0.1909
	<i>OIB Calls</i> _{<i>t</i>-1}	0.0019	0.20	0.8408
	<i>OIB Calls</i> _{<i>t</i>-2}	0.0060	0.65	0.5154
	<i>OIB Calls</i> _{<i>t</i>-3}	-0.0057	-0.60	0.5509
	<i>VIX</i> _{<i>t</i>-1}	0.0470	0.93	0.3550
	<i>VIX</i> _{<i>t</i>-2}	0.0136	0.23	0.8184
	<i>VIX</i> _{<i>t</i>-3}	-0.0299	-0.00	0.5485
Transition probabilities	p_{11}	0.9807		
	p_{22}	0.9469		

Table 6
Sums of VAR Coefficients and Impulse Response Functions
for VAR that includes non-option order imbalances and interaction terms

This table reports results from the following VAR:

$\tilde{\mathbf{x}}_t = \boldsymbol{\alpha} + \sum_{j=1}^3 \boldsymbol{\beta}_j \mathbf{x}_{t-j} + \sum_{j=1}^3 \boldsymbol{\theta}_{1j} OIB\ Calls_{t-j} D_{t-j} + \sum_{j=1}^3 \boldsymbol{\theta}_{2j} OIB\ Puts_{t-j} D_{t-j} + \sum_{j=1}^3 \boldsymbol{\gamma}_j D_{t-j} + \boldsymbol{\varepsilon}_t$, where $\mathbf{x}_t$ is a vector of nine variables (OIB Calls, OIB Puts, OIB E-mini, OIB NYSE, OIB SPY, S&P 500 return, T-bill yield, term spread, and credit spread). In addition to these variables, $\tilde{\mathbf{x}}_t$ includes interaction terms between the call and put option order imbalances and a dummy variable D_t equal to one in weeks when the VIX is in its top quartile and zero otherwise. All variables are measured at weekly intervals. First differences (in basis points) are used for the T-bill yield, the term spread and the credit spread. Panel A shows the sums of VAR coefficients on lags of option order imbalances in the equations shown in column headings. The reported Wald test statistics are for the tests that the corresponding sums of coefficients are equal to zero. Panel B presents the accumulated responses of column variables to Cholesky one-standard-deviation innovations in row variables for a forecast horizon of 10 weeks. The following Cholesky ordering is used: OIB Calls, OIB Calls*High-VIX Dummy, OIB Puts, OIB Puts*High-VIX Dummy, OIB E-mini, OIB NYSE, OIB SPY, S&P 500 Return, T-bill Yield, Term Spread, Credit Spread. Standard errors are shown in parentheses. The sample period is from January 2, 2006 to December 29, 2017 and contains 615 observations, including 156 observations for which the high-VIX dummy is equal to one. Bold text indicates statistical significance at 5% level.

Panel A: Sums of VAR Coefficients

	S&P 500 Return	T-bill Yield	Term Spread	Credit Spread
Sum of coefficients on lags of OIB Calls				
Estimate	-0.0117	0.1146	-0.0707	-0.0233
Wald Chi-square statistic (Sum=0)	0.2457	1.5985	0.3151	0.4010
<i>p</i> -value	0.6201	0.2061	0.5746	0.5266
Sum of coefficients on lags of OIB Calls and interaction of OIB Calls with high-VIX dummy				
Estimate	0.0221	-0.6141	0.2586	0.0350
Wald Chi-square statistic (Sum=0)	0.2169	11.4022	1.0477	0.2248
<i>p</i> -value	0.6414	0.0007	0.3060	0.6354
Sum of coefficients on lags of OIB Puts				
Estimate	0.0138	-0.0624	0.0712	0.0099
Wald Chi-square statistic (Sum=0)	0.2125	0.2962	0.1993	0.0455
<i>p</i> -value	0.6448	0.5863	0.6553	0.8311
Sum of coefficients on lags of OIB Puts and interaction of OIB Puts with high-VIX dummy				
Estimate	0.2863	-0.1015	0.5145	0.0994
Wald Chi-square statistic (Sum=0)	33.8621	0.2905	3.8697	1.6883
<i>p</i> -value	0.0000	0.5899	0.0492	0.1938
Sum of coefficients on lags of OIB E-mini				
Estimate	-0.0822	0.7015	-0.6110	0.1859
Wald Chi-square statistic (Sum=0)	0.3661	1.8231	0.7165	0.7747
<i>p</i> -value	0.5451	0.1770	0.3973	0.3788
Sum of coefficients on lags of OIB NYSE				
Estimate	0.0465	-0.0517	0.1759	0.0266
Wald Chi-square statistic (Sum=0)	1.1363	0.0959	0.5752	0.1539
<i>p</i> -value	0.2864	0.7568	0.4482	0.6949
Sum of coefficients on lags of OIB SPY				
Estimate	-0.0030	-0.2105	0.0791	0.0482
Wald Chi-square statistic (Sum=0)	0.0031	1.0341	0.0757	0.3276
<i>p</i> -value	0.9557	0.3092	0.7832	0.5671

Panel B: Impulse Responses

	S&P 500 Return	T-bill Yield	Term Spread	Credit Spread
OIB Calls	0.10 (0.27)	-0.52 (0.96)	0.43 (1.36)	-0.27 (0.98)
OIB Calls*High-VIX Dummy	0.63 (0.29)	-2.57 (1.02)	2.42 (1.45)	-0.10 (1.05)
OIB Puts	0.50 (0.26)	-0.52 (0.91)	0.45 (1.30)	0.09 (0.95)
OIB Puts*High-VIX Dummy	0.87 (0.26)	0.83 (0.92)	-0.29 (1.29)	-0.65 (0.95)
OIB E-mini	1.28 (0.25)	2.37 (0.90)	0.27 (1.26)	-2.49 (0.93)
OIB NYSE	0.54 (0.29)	0.10 (1.04)	1.31 (1.50)	-0.97 (1.03)
OIB SPY	0.05 (0.22)	-0.93 (0.78)	0.37 (1.10)	1.03 (0.82)

Table 7
Impulse Response Functions for S&P 500 Returns and Macroeconomic Indicators
for VAR that includes SPX, ETF and non-option order imbalances and interaction terms

This table shows accumulated responses of column variables to Cholesky one-standard-deviation innovations in row variables for a forecast horizon of 10 weeks. The VAR specification used to generate the impulse responses is:

$\tilde{\mathbf{x}}_t = \boldsymbol{\alpha} + \sum_{j=1}^3 \boldsymbol{\beta}_j \mathbf{x}_{t-j} + \sum_{j=1}^3 \boldsymbol{\theta}_{1j} OIB\ Calls\ ISE_{t-j} D_{t-j} + \sum_{j=1}^3 \boldsymbol{\theta}_{2j} OIB\ Puts\ ISE_{t-j} D_{t-j} + \sum_{j=1}^3 \boldsymbol{\gamma}_j D_{t-j} + \boldsymbol{\varepsilon}_t$, where $\mathbf{x}_t$ is a vector of 13 variables (OIB Calls ISE, OIB Calls SPX, OIB Calls ISE ETF, OIB Puts ISE, OIB Puts SPX, OIB Puts ISE ETF, OIB E-mini, OIB NYSE, OIB SPY, S&P 500 return, T-bill yield, term spread, and credit spread). In addition to these variables, $\tilde{\mathbf{x}}_t$ includes interaction terms between the ISE index call and put option order imbalances and a dummy variable D_t equal to one in weeks when the VIX is in its top quartile and zero otherwise. All variables are measured at weekly intervals. The following Cholesky ordering is used: OIB Calls ISE, OIB Calls ISE*High-VIX Dummy, OIB Calls SPX, OIB Calls ISE ETF, OIB Puts ISE, OIB Puts ISE*High-VIX Dummy, OIB Puts SPX, OIB Puts ISE ETF, OIB E-mini, OIB NYSE, OIB SPY, S&P 500 Return, T-bill Yield, Term Spread, Credit Spread. The sample period is from January 2, 2006 to December 27, 29, 2017 and contains 615 observations, including 156 observations for which the high-VIX dummy is equal to one. Standard errors are shown in parentheses. Bold text indicates statistical significance at 5% level.

	S&P 500 Return	T-bill Yield	Term Spread	Credit Spread
OIB Calls ISE	0.20 (0.28)	-0.20 (0.97)	0.24 (1.39)	-0.65 (0.99)
OIB Calls ISE*High-VIX Dummy	0.57 (0.30)	-2.70 (1.03)	2.56 (1.47)	0.13 (1.05)
OIB Calls SPX	0.03 (0.26)	1.89 (0.91)	-0.49 (1.30)	-1.53 (0.94)
OIB Calls ISE ETF	-0.11 (0.26)	-0.66 (0.89)	-0.16 (1.27)	0.42 (0.91)
OIB Puts ISE	0.89 (0.26)	-0.12 (0.90)	0.12 (1.30)	-0.41 (0.93)
OIB Puts ISE*High-VIX Dummy	0.89 (0.26)	1.01 (0.91)	-0.28 (1.31)	-0.81 (0.94)
OIB Puts SPX	0.02 (0.30)	-0.10 (1.05)	-1.43 (1.52)	1.25 (1.05)
OIB Puts ISE ETF	-0.28 (0.24)	-2.25 (0.86)	1.30 (1.25)	1.30 (0.86)
OIB E-mini	1.20 (0.25)	1.94 (0.89)	0.78 (1.28)	-2.10 (0.93)
OIB NYSE	0.70 (0.27)	0.62 (0.95)	1.30 (1.38)	-1.73 (0.98)
OIB SPY	-0.01 (0.22)	-0.73 (0.77)	0.09 (1.10)	1.15 (0.81)

Table 8
Sums of VAR Coefficients for Option Order Imbalances
for VAR with interaction terms for weeks of macroeconomic announcements

This table shows sums of VAR coefficients on lags of option order imbalances in the equations shown in column headings. The VAR specification is:

$$\tilde{\mathbf{x}}_t = \boldsymbol{\alpha} + \sum_{j=1}^3 \boldsymbol{\beta}_j \mathbf{x}_{t-j} + \sum_{j=1}^3 \boldsymbol{\theta}_{1j} OIB\ Calls_{t-j} (1 - D_{t-j}) + \sum_{j=1}^3 \boldsymbol{\theta}_{2j} OIB\ Calls_{t-j} D_{t-j} + \sum_{j=1}^3 \boldsymbol{\theta}_{3j} OIB\ Puts_{t-j} (1 - D_{t-j}) + \sum_{j=1}^3 \boldsymbol{\theta}_{4j} OIB\ Puts_{t-j} D_{t-j} + \sum_{j=1}^3 \boldsymbol{\gamma}_j D_{t-j} + \boldsymbol{\varepsilon}_t,$$

where $\mathbf{x}_t$ is a vector of five variables, including the S&P 500 return, T-bill yield, term spread, credit spread, and the VIX. In addition to these variables, vector $\tilde{\mathbf{x}}_t$ contains the four interaction variables (shown on the right-hand-side of the equation) between option order imbalances and a dummy variable, D_t . In Panel A, this dummy is equal to one in weeks that contain the Federal Open Market Committee (FOMC), U.S. Consumer Price Index (CPI), Gross Domestic Product (GDP) and unemployment announcements, and zero otherwise. In Panel B, this dummy is equal to one in weeks preceding these four announcement and zero otherwise. 424 out of 615 weeks in the sample contain at least one of these announcements. All variables are measured at weekly intervals. First differences (in basis points) are used for the T-bill yield, the term spread and the credit spread. The sample period is from January 2, 2006 to December 29, 2017. The reported Wald test statistics are for the tests that the corresponding sums of coefficients are equal to zero. Bold text indicates statistical significance at 5% level.

Panel A. With Dummy Variable for Weeks of Macroeconomic Announcements

	S&P 500 Return	T-bill Yield	Term Spread	Credit Spread	VIX
Sum of coefficients on lags of OIB Calls (weeks without FOMC, CPI, GDP, and unemployment news)					
Estimate	-0.0049	0.0783	0.1454	-0.0110	0.0168
Wald Chi-square statistic (Sum=0)	0.0155	0.2047	0.4410	0.0570	0.1008
p-value	0.9008	0.6510	0.5067	0.8113	0.7509
Sum of coefficients on lags of OIB Calls (weeks with FOMC, CPI, GDP, and unemployment news)					
Estimate	-0.0127	0.0069	-0.1329	0.0022	-0.0116
Wald Chi-square statistic (Sum=0)	0.4209	0.0066	1.0712	0.0044	0.1458
p-value	0.5165	0.9352	0.3007	0.9473	0.7026
Sum of coefficients on lags of OIB Puts (weeks without FOMC, CPI, GDP, and unemployment news)					
Estimate	0.0636	-0.3557	0.3684	-0.0277	-0.0956
Wald Chi-square statistic (Sum=0)	1.3224	4.7350	1.9038	0.1937	1.3997
p-value	0.2502	0.0296	0.1677	0.6499	0.2368
Sum of coefficients on lags of OIB Puts (weeks with FOMC, CPI, GDP, and unemployment news)					
Estimate	0.1067	-0.0028	0.1427	0.0287	-0.1112
Wald Chi-square statistic (Sum=0)	11.5846	0.0008	1.1557	0.4631	8.3169
p-value	0.0007	0.9772	0.2824	0.4962	0.0039

Panel B. With Dummy Variable for Weeks Preceding Macroeconomic Announcements

	S&P 500 Return	T-bill Yield	Term Spread	Credit Spread	VIX
Sum of coefficients on lags of OIB Calls (weeks with preannouncement dummy equal to zero)					
Estimate	-0.0600	0.0261	-0.1470	0.0562	0.0552
Wald Chi-square statistic (Sum=0)	3.1792	0.0287	0.5690	0.9882	1.3242
<i>p</i> -value	0.0746	0.8656	0.4506	0.3202	0.2498
Sum of coefficients on lags of OIB Calls (weeks before FOMC, CPI, GDP, and unemployment news)					
Estimate	0.0095	0.0162	-0.0220	-0.0241	-0.0298
Wald Chi-square statistic (Sum=0)	0.1748	0.0310	0.0308	0.4341	0.8491
<i>p</i> -value	0.6759	0.8603	0.8606	0.5100	0.3568
Sum of coefficients on lags of OIB Puts (weeks with preannouncement dummy equal to zero)					
Estimate	0.0837	-0.0486	0.0578	0.0563	-0.0445
Wald Chi-square statistic (Sum=0)	4.0808	0.0647	0.0608	0.7227	0.6366
<i>p</i> -value	0.0434	0.7992	0.8053	0.3953	0.4249
Sum of coefficients on lags of OIB Puts (weeks before FOMC, CPI, GDP, and unemployment news)					
Estimate	0.1154	-0.1163	0.3388	-0.0347	-0.1599
Wald Chi-square statistic (Sum=0)	6.9261	0.8612	4.0829	0.5142	7.1587
<i>p</i> -value	0.0085	0.3534	0.0433	0.4733	0.0075

Table 9
Sums of VAR Coefficients for Option Order Imbalances
(with positive and negative option order imbalances)

This table shows sums of VAR coefficients on lags of option order imbalances in the equations shown in column headings. The VAR specification is: $\mathbf{x}_t = \boldsymbol{\alpha} + \sum_{j=1}^3 \boldsymbol{\beta}_j \mathbf{x}_{t-j} + \boldsymbol{\varepsilon}_t$, where $\mathbf{x}_t$ is a vector of nine variables (positive OIB Calls, negative OIB Calls, positive OIB Puts, negative OIB Puts, S&P 500 return, T-bill yield, term spread, credit spread, and the VIX). Positive OIB is computed as $\max(0, OIB)$. Negative OIB is computed as $\min(0, OIB)$. All variables are measured at weekly intervals and expressed in percentage terms. The T-bill yield, the term spread and the credit spread are detrended using a quadratic trend. The reported Wald test statistics are for the tests that the corresponding sums of coefficients are equal to zero. The sample period is from January 2, 2006 to December 29, 2017 and contains 615 observations. Bold text indicates statistical significance at 5% level.

	S&P 500 Return	T-bill Yield	Term Spread	Credit Spread	VIX
Sum of coefficients on lags of Positive OIB Calls	-0.0852	-0.0459	-0.0663	-0.0095	0.0665
Wald Chi-square statistic (Sum=0)	3.0586	0.0590	0.0671	0.0165	0.9614
<i>p</i> -value	0.0803	0.8081	0.7957	0.8978	0.3268
Sum of coefficients on lags of Negative OIB Calls	0.0213	0.0456	-0.0335	-0.0012	-0.0297
Wald Chi-square statistic (Sum=0)	0.4717	0.1427	0.0422	0.0007	0.4703
<i>p</i> -value	0.4922	0.7056	0.8373	0.9791	0.4929
Sum of coefficients on lags of Positive OIB Puts	0.1342	-0.1087	0.2619	0.0615	-0.1630
Wald Chi-square statistic (Sum=0)	7.4402	0.3238	1.0259	0.6733	5.6585
<i>p</i> -value	0.0064	0.5694	0.3111	0.4119	0.0174
Sum of coefficients on lags of Negative OIB Puts	0.0639	-0.0681	0.1524	-0.0198	-0.0607
Wald Chi-square statistic (Sum=0)	2.1733	0.1637	0.4471	0.0896	1.0105
<i>p</i> -value	0.1404	0.6858	0.5037	0.7646	0.3148

Table 10
Average Bid-Ask Spread and Trading Volume for Index Call and Put Options

This table shows the average bid-ask spread and weekly trading volume. The bid-ask spread is computed as the difference between the best closing ask and the best closing bid quotes, divided by the quote midpoint. The bid-ask spreads are expressed in percentage terms. The average bid-ask spread is computed as a simple average of volume-weighted average spreads for each week. Option contracts on the following two indexes are included: Russell 2000 Index (RUT) and Nasdaq-100 Index (NDX). Options with bid or ask quotes less than or equal to zero, those with zero or negative bid-ask spread, those with the bid-ask spread above 50% and those with open interest equal to zero are removed from the sample. Panel C compares the average bid-ask spreads for call and put options during weeks with and without FOMC, CPI, GDP, and unemployment announcements. The sample period is from January 2, 2006 to December 29, 2017 and contains 615 weekly observations. Bold text indicates statistical significance at 5% level.

	Full sample	VIX below top quartile	VIX in top quartile	<i>t</i> -stat for difference	<i>p</i> -value for difference
Panel A. Average bid-ask spread					
Call Options	12.32	12.47	11.86	2.36	0.0191
Put Options	11.15	11.73	9.44	9.88	0.0000
Difference between calls and puts	1.17	0.74	2.42	-5.62	0.0000
<i>t</i> -statistic for difference	9.33	5.46	9.14		
<i>p</i> -value	0.0000	0.0000	0.0000		
Panel B. Average trading volume					
Call Options	204,085	180,578	273,248	-8.61	0.0000
Put Options	287,787	259,794	370,149	-7.38	0.0000
Difference between calls and puts	-83,702	-79,216	-96,901	1.77	0.0791
<i>t</i> -statistic for difference	-23.44	-24.38	-10.22		
<i>p</i> -value	0.0000	0.0000	0.0000		

Panel C. Average bid-ask spread in weeks with and without major macroeconomic announcements

	Weeks without major macro news (<i>N</i> = 191)	Weeks with major macro news (<i>N</i> = 424)	<i>t</i> -stat for difference	<i>p</i> -value for difference
Call Options	12.29	12.33	-0.18	0.8590
Put Options	11.57	10.96	2.31	0.0216

Figure 1
Order Imbalances and S&P 500 Returns

This figure shows weekly time series of order imbalances and S&P 500 returns over the period from January 2, 2006 to December 29, 2017.

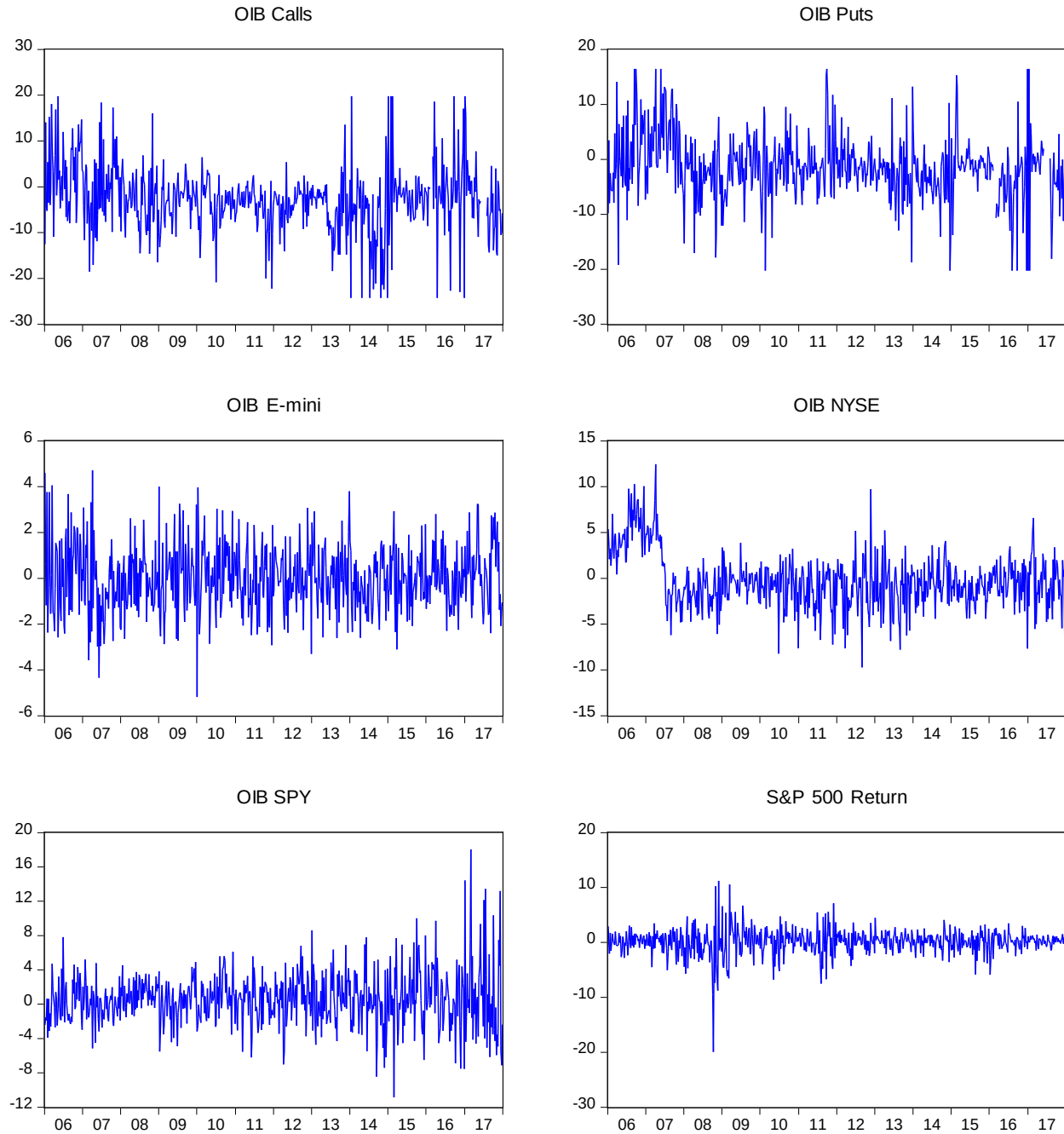


Figure 2
Relation between Put Option Order Imbalances and S&P 500 Returns

This figure shows the relation between the weekly index put option order imbalance sorted into quintiles and the S&P 500 index returns over the following one and two weeks. The sample period is from January 2, 2006 to December 29, 2017.

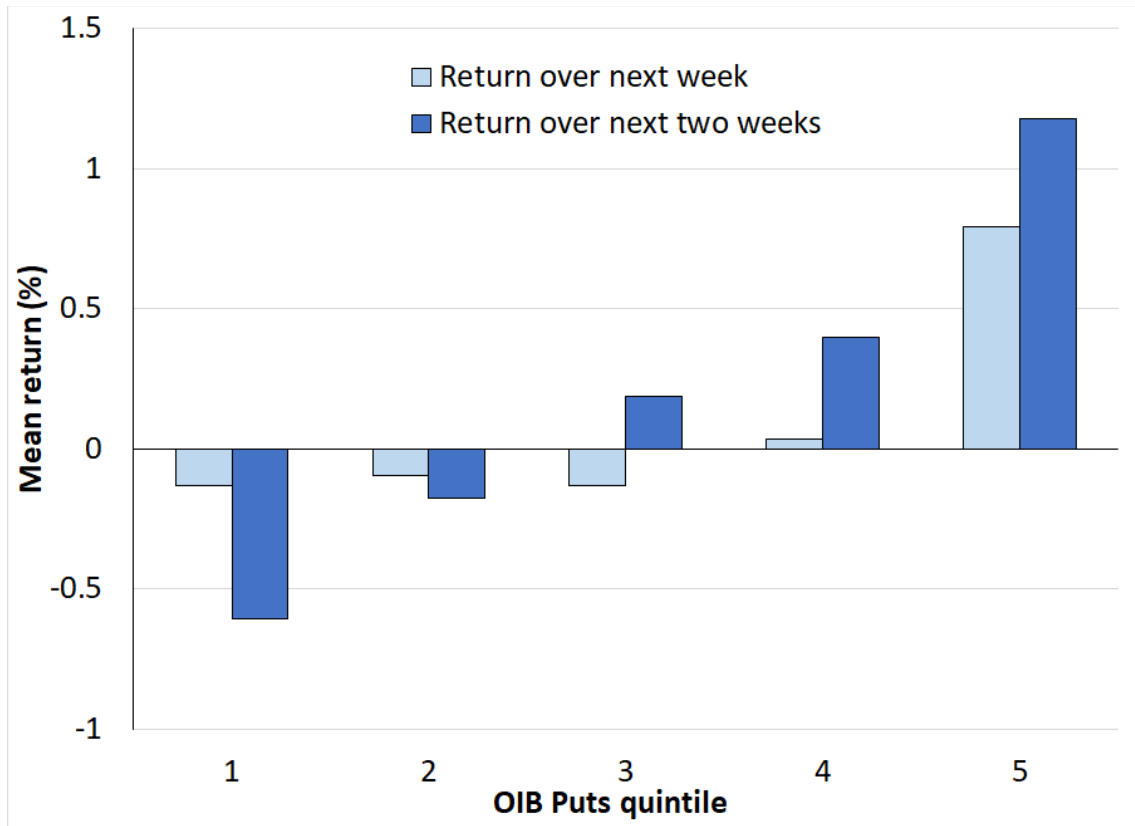


Figure 3
Impulse Response Functions for S&P 500 Returns and Macroeconomic Indicators

This figure shows accumulated impulse response functions from the following VAR:

$\mathbf{x}_t = \boldsymbol{\alpha} + \sum_{j=1}^3 \boldsymbol{\beta}_j \mathbf{x}_{t-j} + \boldsymbol{\varepsilon}_t$, where $\mathbf{x}_t$ is a vector of seven variables (OIB Calls, OIB Puts, S&P 500 return, T-bill yield, term spread, credit spread, and the VIX). All variables are measured at weekly intervals. The following Cholesky ordering is used: OIB Calls, OIB Puts, S&P 500 Return, T-bill Yield, Term Spread, Credit Spread, VIX. The sample period is from January 2, 2006 to December 29, 2017. Time on the horizontal axis is in weeks.

Accumulated Response to Cholesky One S.D. Innovations ± 2 S.E.

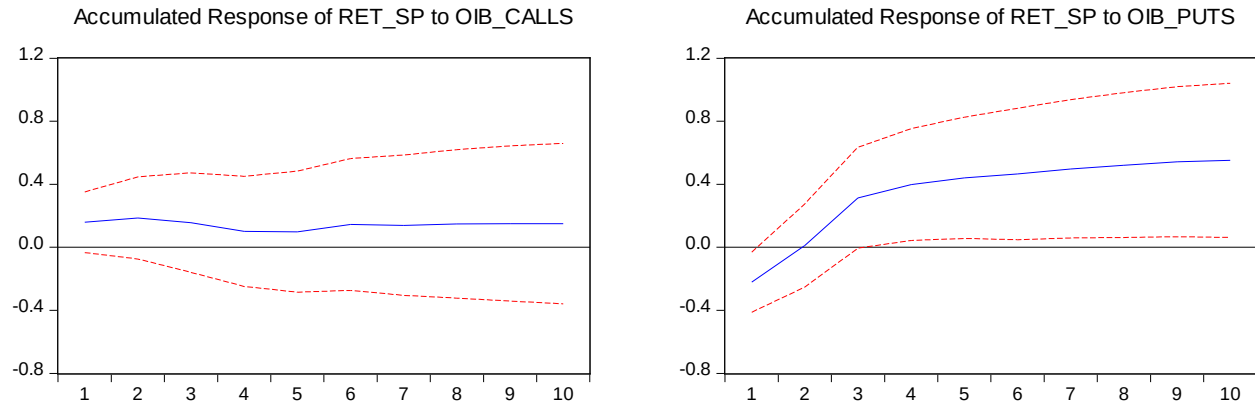


Figure 4
Impulse Response Functions for S&P 500 Returns and Macroeconomic Indicators
(for VAR with interaction terms for high VIX periods)

This figure shows accumulated impulse response functions from the following VAR:

$\tilde{\mathbf{x}}_t = \boldsymbol{\alpha} + \sum_{j=1}^3 \boldsymbol{\beta}_j \mathbf{x}_{t-j} + \sum_{j=1}^3 \boldsymbol{\theta}_{1j} OIB\ Calls_{t-j} D_{t-j} + \sum_{j=1}^3 \boldsymbol{\theta}_{2j} OIB\ Puts_{t-j} D_{t-j} + \sum_{j=1}^3 \boldsymbol{\gamma}_j D_{t-j} + \boldsymbol{\varepsilon}_t$, where $\mathbf{x}_t$ is a vector of six variables (OIB Calls, OIB Puts, S&P 500 return, T-bill yield, term spread, and credit spread). In addition to these variables, $\tilde{\mathbf{x}}_t$ includes interaction terms between the call and put option order imbalances and a dummy variable D_t equal to one in weeks when the VIX is in its top quartile and zero otherwise. All variables are measured at weekly intervals. The following Cholesky ordering is used: OIB Calls, OIB Calls*High-VIX Dummy, OIB Puts, OIB Puts*High-VIX Dummy, S&P 500 Return, T-bill Yield, Term Spread, Credit Spread. The sample period is from January 2, 2006 to December 29, 2017 and contains 615 observations, including 156 observations for which the high-VIX dummy is equal to one. Time on the horizontal axis is in weeks.

Accumulated Response to Cholesky One S.D. Innovations ± 2 S.E.

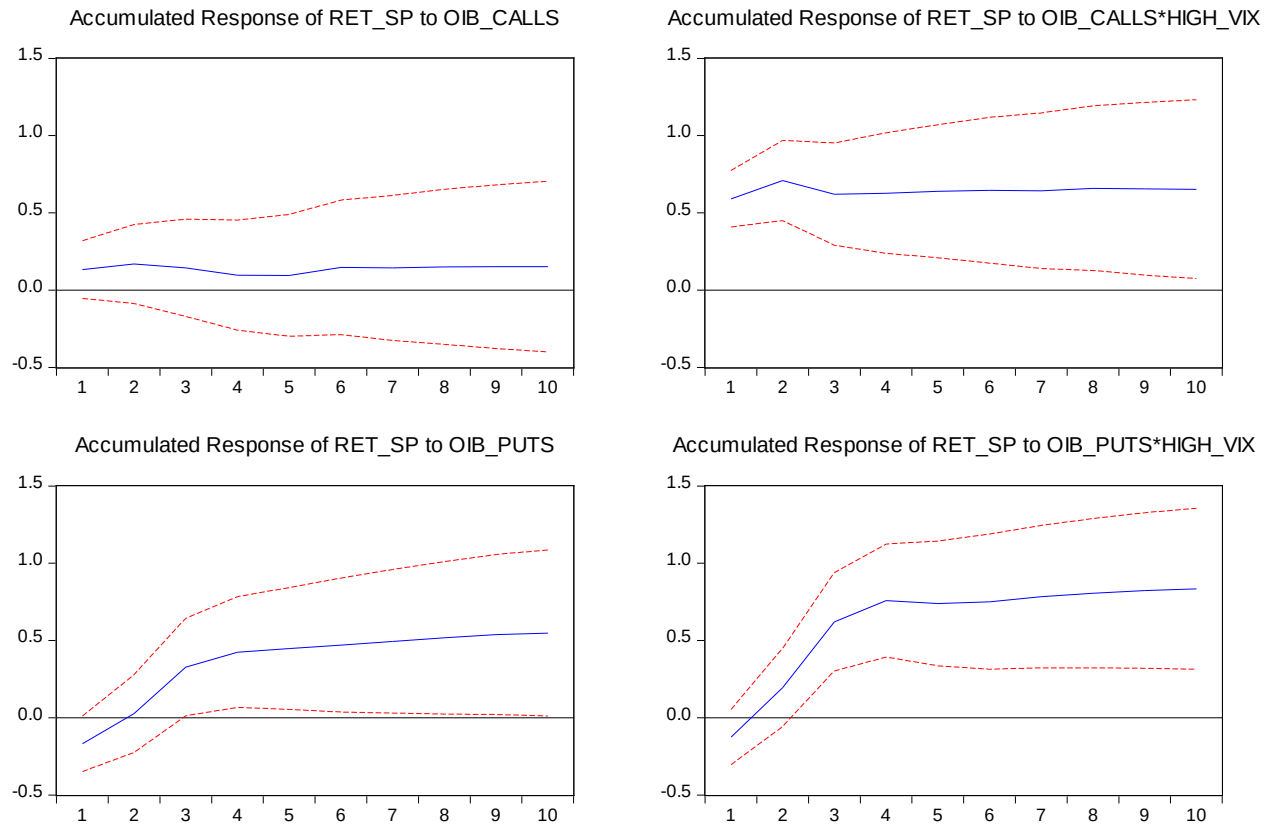
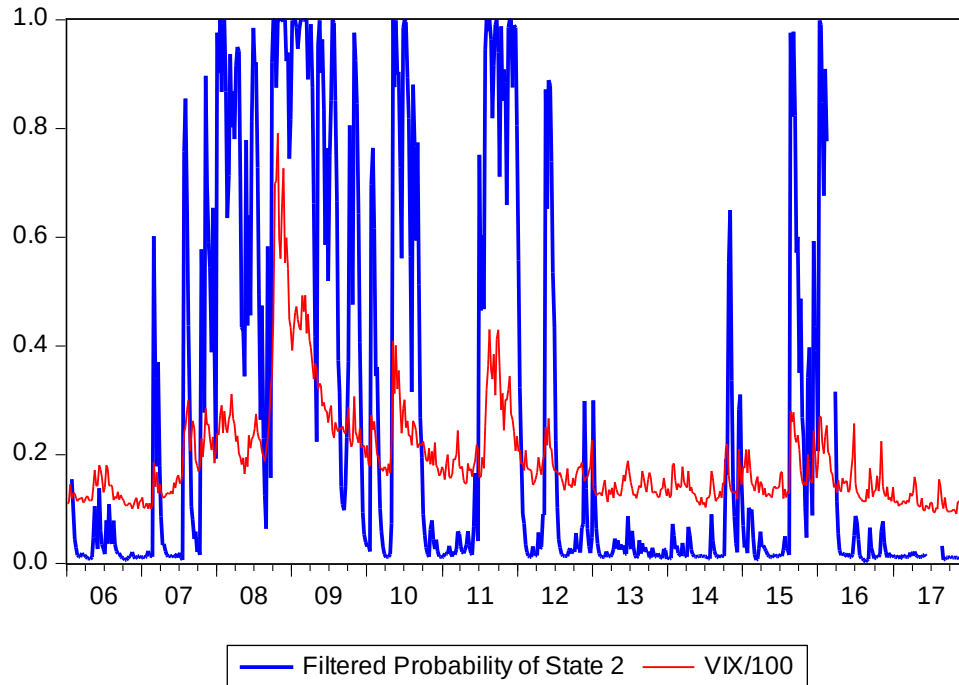


Figure 5
Filtered Probability of State 2 (high return variance)

This figure shows the filtered probability of State 2, characterized by high return variance, obtained from the following Markov-switching model:

$R_t = \alpha + \sum_{j=1}^3 \theta_j R_{t-j} + \sum_{j=1}^3 \beta_j OIB\ Calls_{t-j} + \sum_{j=1}^3 \gamma_j VIX_{t-j} + \sum_{j=1}^3 \delta_{j,s_t} OIB\ Puts_{t-j} + \varepsilon_t$, where R_t is the S&P 500 index return, $\varepsilon_t \sim N(0, \sigma_{\varepsilon_t}^2)$, and the unobserved state variable $s_t = \{1, 2\}$ follows a Markov process with fixed transition probabilities. Also shown is the VIX level. All variables are measured at weekly intervals. The sample period is from January 2, 2006 to December 29, 2017 and contains 615 observations.



Appendix A

Brief Background on Markov-Switching Models

Suppose that variable y_t follows the following process:

$$y_t = x_t' \beta_{s_t} + \varepsilon_t,$$
$$\varepsilon_t \sim N(0, \sigma_{s_t}^2),$$

where x_t is a vector of regressors, β_{s_t} is a vector of coefficients and $s_t = \{1, 2\}$ is an unobserved state variable that follows a first-order Markov process with fixed transition probabilities:

$$P(s_t = 1 | s_{t-1} = 1) = p_{11} = \frac{\exp(q_1)}{1 + \exp(q_1)},$$
$$P(s_t = 2 | s_{t-1} = 2) = p_{22} = \frac{\exp(q_2)}{1 + \exp(q_2)}.$$

Because the state probabilities depend on their own previous values, the parameter estimation must be performed recursively. In the first step, the one-period ahead state probabilities are formed as follows:

$$P(s_t = m | \Psi_{t-1}) = \sum_{j=1}^2 P(s_t = m | s_{t-1} = j) P(s_{t-1} = j | \Psi_{t-1}),$$

where Ψ_{t-1} is the information set in period $t - 1$. These probabilities are used to form the joint densities of y_t and s_t :

$$f(y_t, s_t = m | \Psi_{t-1}) = \frac{1}{\sqrt{2\pi\sigma_m^2}} \exp\left(\frac{-(y_t - x_t' \beta_m)^2}{2\sigma_m^2}\right) P(s_t = m | \Psi_{t-1}).$$

The likelihood contribution for a given period is then obtained as:

$$L_t = \sum_{j=1}^2 f(y_t, s_t = j | \Psi_{t-1}).$$

The full likelihood function is obtained by summing the likelihood contributions across periods:

$$\ln L = \sum_{t=1}^T \ln L_t.$$

The parameters $(\beta_1, \beta_2, \sigma_1, \sigma_2, q_1, q_2)$ are estimated by maximum likelihood. In the final step, the state probabilities are smoothed using the full sample of data.

Internet Appendix to Accompany “Index Option Trading Activity and Market Returns”

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A. Out-of-sample Forecasting Test

Goyal and Welch (2008) show that predictors of the equity premium perform poorly in out-of-sample tests. Following Goyal and Welch (2008) and Campbell and Thompson (2008), we use the following out-of-sample R^2 statistic to evaluate the out-of-sample performance of a predictive regression that uses the order imbalance of index put options:

$$R_{OS}^2 = 1 - \frac{\sum_{t=T-P}^{T-1} (R_{t+1} - \hat{R}_{t+1})^2}{\sum_{t=T-P}^{T-1} (R_{t+1} - \bar{R}_{t+1})^2},$$

where T is the overall sample size, P is the number of observations in the out-of-sample period used for forecast evaluation, $\hat{R}_{t+1}$ is the one-period-ahead forecast generated by a predictive regression, and $\bar{R}_{t+1}$ is the recursively estimated historical average. The R_{OS}^2 measures the proportional reduction in the mean square prediction error (MSPE) for the predictive regression relative to the historical mean benchmark. Our predictive regression is:

$$R_{t+1} = \alpha + \sum_{j=0}^2 \theta_j R_{t-j} + \sum_{j=0}^2 \beta_j OIB\ Puts_{t-j} + \sum_{j=0}^2 \partial_j OIB\ Puts_{t-j} D_{t-j} + \sum_{j=0}^2 \gamma_j D_{t-j} + \varepsilon_{t+1},$$

where R_t is the S&P 500 return and D_t is equal to one in weeks when the VIX is in its top quartile and zero otherwise.¹ The regression is estimated using a recursive (expanding) estimation window. To make sure that the results are robust to the choice of the estimation window, we use 2010, 2011, 2012, 2013, and 2014 as the starting points of the out-of-sample forecasting period.

Clark and West (2007) show that when one compares predictive accuracy of a simple model with that of a larger model that nests the simple model, the MSPE of the larger model is expected to be larger than the MSPE of the simple model under the null. This happens because the larger model introduces noise into its forecasts by adding superfluous parameters. If one does

¹ The top quartile of VIX in the predictive regression is based on the data from January 1, 1990 to day t .

not adjust for this estimation noise, tests of predictability based on forecast errors of the two models are biased against finding predictability. Clark and West (2007) show how to adjust for this bias and propose a simple “MSPE-adjusted” statistic to test for equal predictive accuracy of nested models. They show that the MSPE-adjusted statistic is approximately standard normal and recommend using the usual standard normal critical values for one-tailed tests. To compute the MSPE-adjusted statistic, we follow Clark and West (2007) and define:

$$f_{t+1} = (R_{t+1} - \bar{R}_{t+1})^2 - \left[(R_{t+1} - \hat{R}_{t+1})^2 - (\bar{R}_{t+1} - \hat{R}_{t+1})^2 \right].$$

The test for equal MSPEs is conducted by regressing $\{f_{t+1}\}_{T-P}^{T-1}$ on a constant. The MSPE-adjusted statistic is the t -statistic of the resulting coefficient, and the one-tailed (upper-tail) test is based on the standard normal critical values.²

Campbell and Thompson (2008, footnote 5) discuss the finding of Clark and West (2007) that the MSPEs of predictive regressions are biased upward. They note that, due to this bias, the expected R_{OS}^2 under the null of no predictability is negative, and a zero R_{OS}^2 can be viewed as weak evidence of predictability. Since we want to test whether the stock market returns are predictable using order imbalances of index puts, we compute the CW-adjusted R_{OS}^2 that adjusts for the bias in the MSPE of the predictive regression:³

$$\text{CW-adjusted } R_{OS}^2 = 1 - \frac{\sum_{t=T-P}^{T-1} \left[(R_{t+1} - \hat{R}_{t+1})^2 - (\bar{R}_{t+1} - \hat{R}_{t+1})^2 \right]}{\sum_{t=T-P}^{T-1} (R_{t+1} - \bar{R}_{t+1})^2}.$$

The results reported in Panel A of Table A1 show that the estimates of R_{OS}^2 are positive and the CW-adjusted R_{OS}^2 estimates are sizable for all out-of-sample forecasting periods

² The MSPE-adjusted statistic is used, for example, by Rapach, Strauss and Zhou (2010) and Rapach, Ringgenberg and Zhou (2016).

³ This statistic is proposed by Hillebrand, Lee and Medeiros (2014).

mentioned above.⁴ In each of these cases, the null hypothesis of no predictability is rejected at least at the 10% significance level based on the MSPE-adjusted statistic. These results strong compared to other known market return predictors, which often perform remarkably poorly in out-of-sample tests as Goyal and Welch (2008) show.

Campbell and Thompson (2008) argue that if one assumes that investors rule out a negative equity premium, forecasts from predictive regressions can be set to zero when they are negative.⁵ Panel B of Table A1 shows out-of-sample forecasting results with the S&P 500 return forecasts restricted to be non-negative for both the predictive regression and the historical mean benchmark. These results are generally similar to those based on unconstrained forecasts, although the R_{OS}^2 estimates are more stable across the out-of-sample forecasting periods. Overall, the out-of-sample forecasting test provides evidence that the order imbalance of index put options traded on the ISE predicts the S&P 500 index returns out-of-sample. Note that we are not arguing that put option order imbalance can be used as a device to time entries into or exits from the stock index; instead we are simply showing that put option imbalance predicts market returns.

⁴ For comparison, Rapach, Ringgenberg and Zhou (2016) and Martin (2017) report R_{OS}^2 statistics of about 1.94% and 0.42%, respectively, for one-month-ahead forecasts of the equity premium.

⁵ Pettenuzzo, Timmermann and Valkanov (2014) impose a similar constraint on forecasts of the equity premium.

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Table A1
Out-of-sample Forecasting Results

This table reports the Campbell and Thompson (2008) out-of-sample R^2 statistic (R_{OS}^2) for a predictive regression for the S&P 500 return. R_{OS}^2 measures the proportional reduction in the mean square prediction error (MSPE) for the following predictive regression relative to the historical mean benchmark. The predictive regression is: $R_{t+1} = \alpha + \sum_{j=0}^2 \theta_j R_{t-j} + \sum_{j=0}^2 \beta_j OIB Puts_{t-j} + \sum_{j=0}^2 \partial_j OIB Puts_{t-j} D_{t-j} + \sum_{j=0}^2 \gamma_j D_{t-j} + \varepsilon_{t+1}$, where R_t is the S&P 500 return, $OIB Puts_t$ is order imbalance for index put options, and D_t is equal to one in weeks when the VIX is in its top quartile and zero otherwise. The table also shows the CW-adjusted R_{OS}^2 proposed by Lee, Hillebrand and Medeiros (2014) and the Clark and West (2007) MSPE-adjusted statistic for testing the null hypothesis that the MSPE of the historical mean is less than or equal to the MSPE of the predictive regression against the alternative hypothesis that the MSPE of the historical mean is greater than the MSPE of the predictive model. The statistics in Panel A are computed using unconstrained forecasts. The statistics in Panel B are computed using forecasts for both the historical mean benchmark and the predictive regression restricted to be non-negative. The full sample period is from January 2, 2006 to December 29, 2017 and contains 615 observations. The out-of-sample forecasting period is shown in the first column of the table. *, **, *** indicate statistical significance at 10%, 5%, and 1% levels, respectively, based on a one-tailed test.

Out-of-sample period	R_{OS}^2 (%)	CW-adjusted R_{OS}^2 (%)	MSPE-adjusted statistic
Panel A. Unconstrained forecasts			
January 2010 – December 2017	0.89	9.19	1.94**
January 2011 – December 2017	2.20	10.29	1.81**
January 2012 – December 2017	1.20	4.14	1.91**
January 2013 – December 2017	0.41	3.39	1.48*
January 2014 – December 2017	0.95	3.97	1.52*
Panel B. Non-negative forecasts			
January 2010 – December 2017	2.19	7.87	1.95**
January 2011 – December 2017	2.65	8.57	1.76**
January 2012 – December 2017	2.20	3.45	2.12**
January 2013 – December 2017	1.69	2.84	1.80**
January 2014 – December 2017	1.54	2.76	1.55*

B. Return Predictability Conditional on Order Type

In the table below, we show the predictive power of options order flows separately from small customers:

Table A2
Impulse Response Functions for S&P 500 Returns and Macroeconomic Indicators
VAR with order imbalances of small customer and other trades

This table shows accumulated responses of column variables to Cholesky one-standard-deviation innovations in row variables for a forecast horizon of 10 weeks. The VAR specification used to generate the impulse responses is: $\mathbf{x}_t = \boldsymbol{\alpha} + \sum_{j=1}^3 \boldsymbol{\beta}_j \mathbf{x}_{t-j} + \boldsymbol{\varepsilon}_t$, where $\boldsymbol{\alpha}$ is a vector of constant terms, $\boldsymbol{\beta}_j$ is the vector of coefficients for lag j , $\mathbf{x}_t$ is a vector of nine variables (OIB Calls Small, OIB Calls Other, OIB Puts Small, OIB Puts Other, S&P 500 return, T-bill yield, term spread, credit spread, and the VIX), and $\boldsymbol{\varepsilon}_t$ is a vector of random disturbances. OIB Calls Small and OIB Puts Small are the order imbalances of customer trades not exceeding 100 contracts per trade. OIB Calls Other and OIB Puts Other are the order imbalances of larger customer trades and firm trades. The option order imbalances are computed by dividing the difference between the corresponding buy and sell volumes by the total volume of customer and firm trades. All variables are measured at weekly intervals. All variables except the T-bill yield, the term spread and the credit spread are expressed in percentage terms. First differences (in basis points) are used for the T-bill yield, the term spread and the credit spread. The following Cholesky ordering is used: OIB Calls Small, OIB Calls Other, OIB Puts Small, OIB Puts Other, S&P 500 Return, T-bill Yield, Term Spread, Credit Spread, VIX. Standard errors are shown in parentheses. The sample period is from January 2, 2006 to December 29, 2017 and contains 615 observations. Bold text indicates statistical significance at 5% level.

	S&P 500 Return	T-bill Yield	Term Spread	Credit Spread	VIX
OIB Calls Small (small customer trades)	0.31 (0.26)	0.82 (1.09)	0.12 (1.33)	-0.63 (0.96)	-2.22 (1.89)
OIB Calls Other (firm and larger customer trades)	-0.12 (0.27)	-0.94 (1.12)	0.40 (1.38)	-0.07 (0.98)	0.69 (1.90)
OIB Puts Small (small customer trades)	0.56 (0.25)	-0.29 (1.06)	-0.44 (1.29)	0.05 (0.95)	-2.76 (1.88)
OIB Puts Other (firm and larger customer trades)	0.13 (0.25)	-0.24 (1.05)	1.87 (1.27)	-0.45 (0.93)	0.32 (1.86)

C. VAR with Non-option Order Imbalances

The figure below provides impulse responses for the VAR with non-option order imbalances and interaction terms for option order imbalances.

Figure A1

Impulse Response Functions for S&P 500 Returns and Macroeconomic Indicators

This figure shows accumulated impulse response functions from the following VAR:

$\tilde{\mathbf{x}}_t = \alpha + \sum_{j=1}^3 \beta_j \mathbf{x}_{t-j} + \sum_{j=1}^3 \theta_{1j} OIB\ Calls_{t-j} D_{t-j} + \sum_{j=1}^3 \theta_{2j} OIB\ Puts_{t-j} D_{t-j} + \sum_{j=1}^3 \gamma_j D_{t-j} + \varepsilon_t$, where $\mathbf{x}_t$ is a vector of nine variables (OIB Calls, OIB Puts, OIB E-mini, OIB NYSE, OIB SPY, S&P 500 return, T-bill yield, term spread, and credit spread). In addition to these variables, $\tilde{\mathbf{x}}_t$ includes interaction terms between the call and put option order imbalances and a dummy variable D_t equal to one in weeks when the VIX is in its top quartile and zero otherwise. All variables are measured at weekly intervals. The following Cholesky ordering is used: OIB Calls, OIB Calls*High-VIX Dummy, OIB Puts, OIB Puts*High-VIX Dummy, OIB E-mini, OIB NYSE, OIB SPY, S&P 500 Return, T-bill Yield, Term Spread, Credit Spread. The sample period is from January 2, 2006 to December 29, 2017 and contains 615 observations, including 156 observations for which the high-VIX dummy is equal to one. Time on the horizontal axis is in weeks.

Accumulated Response to Cholesky One S.D. Innovations ± 2 S.E.

